

POWER CONVEXITY OF SOLUTIONS TO THE COMPLEX MONGE-AMPÈRE EQUATION IN $\mathbb{C}^2$

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ABSTRACT. We establish the power convexity of the solution to the Dirichlet problem for the complex Monge-Ampère equation on bounded, smooth, strictly convex domains in $\mathbb{C}^2$. More precisely, if u solves $\det(u_{i\bar{j}}) = 1$ with zero boundary data, then we prove that $v = -\sqrt{-u/2}$ is strictly convex. Our proof is based on a constant rank theorem for the real Hessian $\nabla^2 v$ and a deformation argument. A main difficulty is that the standard Hessian-based auxiliary functions are not invariant under the complex linear coordinate changes required in our analysis. To overcome this obstacle, we introduce an auxiliary matrix K and construct invariant test functions from its trace and determinant. By deriving separate differential inequalities for the cases where $\nabla^2 v$ has minimal rank 2 and 3, respectively, we obtain the constant rank theorem and hence the desired strict convexity.

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1. INTRODUCTION

1.1. Main result. In 1971, Makar-Limanov [40] studied the Dirichlet problem for the Laplace equation

$$(1.1) \quad \begin{cases} \Delta u = -1 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

where $\Omega \subset \mathbb{R}^2$ is a bounded, strictly convex domain with smooth boundary. He proved that the function $\sqrt{u}$ is strictly concave in Ω . Recall that the unique solution u of the Dirichlet problem (1.1) is commonly referred to as the torsion function of Ω . That is to say, Makar-Limanov [40] established the 1/2-concavity of the torsion function of a planar bounded convex domain. Since then, many authors have investigated the convexity of solutions to various kinds of elliptic partial differential equations (PDEs); see Section 1.2 below.

In this work, we investigate the convexity of solutions of the Dirichlet problem for the complex Monge-Ampère equation. The complex Monge-Ampère equation is a fully nonlinear elliptic PDE of the form

$$\det(u_{i\bar{j}}) = f,$$

where u is a plurisubharmonic function, and $(u_{i\bar{j}})$ is its complex Hessian in local holomorphic coordinates. This equation plays a fundamental role in complex geometry and fully nonlinear PDE theory.

Let us first recall the solvability of the Dirichlet problem for the complex Monge-Ampère equation on a bounded domain in $\mathbb{C}^n$. Assume that $\Omega \subset \mathbb{C}^n$ is a bounded strongly pseudoconvex domain with smooth boundary, and that $0 < f \in C^\infty(\bar{\Omega})$, $\varphi \in C^\infty(\partial\Omega)$. Then Caffarelli, Kohn, Nirenberg and Spruck [12] proved that the Dirichlet problem

$$(1.2) \quad \begin{cases} \det(u_{i\bar{j}}) = f & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega, \end{cases}$$

admits a unique strictly plurisubharmonic solution $u \in C^\infty(\bar{\Omega})$.

Our goal in this paper is to investigate the power convexity of the unique solution u to problem (1.2) when $f = 1$, $\varphi = 0$ and $n = 2$. Precisely, our main result is as follows.

Theorem 1.1. *Let $\Omega \subset \mathbb{C}^2$ be a bounded, strictly convex domain with smooth boundary. Suppose that $u \in C^\infty(\bar{\Omega})$ is the unique solution of the Dirichlet problem*

$$(1.3) \quad \begin{cases} \det(u_{i\bar{j}}) = 1 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Then the function $v = -\sqrt{-u/2}$ is strictly convex in Ω .

Remark 1.2. *To the best of our knowledge, at the time of its initial announcement, this is the first known result addressing the power convexity of solutions to complex Monge-Ampère equations. For $n \geq 3$, it remains an open question whether the unique solution to the Dirichlet problem (1.3) enjoys the same power convexity property. We note that such a higher-dimensional generalization may fail to hold.*

1.2. Previous works. Convexity has long been a fundamental and intriguing topic in the study of elliptic PDEs. Over the decades, many authors have made significant contributions to this field. Below, we provide a brief overview of the historical developments on the convexity of solutions to elliptic PDEs.

As we mentioned earlier, the $1/2$ -concavity of the two-dimensional torsion function was first studied by Makar-Limanov [40]. His key idea was to introduce the auxiliary function

$$P = v^4 \det(\nabla^2 v),$$

where $v = \sqrt{u}$ and u satisfies the Dirichlet problem (1.1). He then showed that P is superharmonic in Ω . In 1976, Brascamp and Lieb [8] proved the log-concavity of the first Dirichlet eigenfunction of the Laplacian of a bounded convex domain using a parabolic method. Acker, Payne and Philippin [1] later provided an alternative proof of Brascamp and Lieb's result following the approach of Makar-Limanov [40] for the two-dimensional case.

In the early 1980s, Korevaar [32, 33] made a breakthrough in studying the convexity of solutions to linear and quasilinear elliptic equations. Specifically, for a bounded convex domain $\Omega \subset \mathbb{R}^n$ and a solution u , he introduced the convexity test function

$$\mathcal{C}(x, y, \lambda) = u((1 - \lambda)x + \lambda y) - (1 - \lambda)u(x) - \lambda u(y)$$

for $(x, y, \lambda) \in \bar{\Omega} \times \bar{\Omega} \times [0, 1]$. Clearly, the nonnegativity of $\mathcal{C}(x, y, \lambda)$ is equivalent to the concavity of u . This method is commonly known as Korevaar's concavity maximum principle. Using this, Korevaar reproved the result of Brascamp and Lieb [8] (see also Caffarelli and Spruck [13]). Later, Kennington [31] and Kawohl [30] presented an improved version that could be used to establish the $1/2$ -concavity of the torsion function in high dimensions. For problems involving the p -Laplacian or the Finsler Laplacian, due to the lack of regularity in solutions, Korevaar's concavity maximum principle cannot be applied directly. However, by suitable approximation arguments, power convexity results have been derived

in [7, 41, 43], among others. In 2021, Langford and Scheuer [36] extended Korevaar's concavity maximum principle to spherical settings, and proved the convexity of solutions to certain fully nonlinear degenerate elliptic equations under appropriate conditions.

In 1997, Alvarez, Lasry and Lions [2] developed a new method, the convex envelope method, to establish the convexity of viscosity solutions of general fully nonlinear degenerate elliptic equations

$$F(\nabla^2 v, \nabla v, v, x) = 0$$

in a bounded convex domain Ω of $\mathbb{R}^n$. In particular, they gave an alternative proof of previous results, including the 1/2-concavity of the torsion function and the log-concavity of the first Dirichlet eigenfunction of the Laplacian. This method was also extended to handle the power convexity of certain elliptic boundary value problems directly (without applying a power transformation) as well as the power convexity of solutions of semilinear elliptic systems in [6, 27]. Recently, another application of the convex envelope method was given by Crasta and Fragalà [19], who proved that the unique viscosity solution u to the Dirichlet problem

$$\begin{cases} \Delta_\infty u = -1 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

is 3/4-concave and in $C^1(\Omega)$. Here Δ_∞ denotes the usual infinity Laplacian.

Another powerful tool for producing convex solutions is the so-called microscopic convexity principle (also called constant rank theorem). This approach was first discovered by Caffarelli and Friedman [10] for semilinear elliptic equations in the two-dimensional case. Combining the deformation process, they established the 1/2-concavity of the torsion function and the log-concavity of the first Dirichlet eigenfunction of the Laplacian in dimension two. Later, Korevaar and Lewis [34] proved the analogous results in high dimensions. A similar idea also appeared at the same time in Singer, Wong, Yau and Yau [46] to deal with the log-concavity of the first Dirichlet eigenfunction of the Laplacian.

For fully nonlinear elliptic equations, the constant rank theorem remains highly effective in dealing with problems related to convexity. One of the earliest results in this direction was obtained by Ma and Xu [39], who studied the Dirichlet problem

$$\begin{cases} \sigma_2(\nabla^2 u) = 1 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

where $\Omega \subset \mathbb{R}^3$ is a bounded, convex domain with smooth boundary, and demonstrated that $-\sqrt{-u}$ is strictly convex in Ω . In the same way, Liu, Ma and Xu [37] investigated the log-concavity of the eigenfunction of the 2-Hessian operator and derived a Brunn-Minkowski inequality for the 2-Hessian eigenvalue. See Salani [44] for new proofs of the power convexity results in [37, 39] as well as discussions on Brunn-Minkowski inequalities of the 2-torsion functional. Much more recently, several lower-dimensional convexity results

of solutions to fully nonlinear elliptic boundary value problems appeared via constant rank theorems, for example [14, 26, 49]. For historical developments on constant rank theorems of convex solutions of fully nonlinear elliptic PDEs and their applications in geometry, one can see [3, 4, 9, 11, 42, 47, 48], [16, 18, 20, 21, 22], and the references therein.

Like the concavity maximum principle, the convex envelope and the constant rank theorem, Makar-Limanov's approach has also been extended to high dimensions to deal with $1/2$ -concavity of the torsion function and the log-concavity of the first Dirichlet eigenfunction of the Laplacian; see [28, 29, 38]. There are also many other interesting results on convexity of solutions of elliptic PDEs. For example, it is worth mentioning that Kulczycki [35] drew inspiration from the constant rank theorem and showed the concavity of the solution to the Dirichlet problem for the $1/2$ -Laplacian in dimension two. In the complex setting, Hu [23] proved that the complex Hessian of the solution to the Dirichlet problem for the homogeneous complex Monge-Ampère equation on a ring-shaped domain in $\mathbb{C}^n$ has constant rank $n - 1$ in the weak sense. More recently, Hu and Sheng [25] established the strict convexity of the potential function of a complete Kähler-Einstein metric on a bounded strictly convex domain in $\mathbb{C}^n$.

Remark 1.3. *After the completion of the present work [50], we became aware of two recent related preprints. Chen, Hu and Sheng [17] obtained the same power convexity result in complex dimension two by a different approach, based on estimates for the smallest eigenvalue of an auxiliary matrix using complex derivatives. Chen, Li and Ma [15] established strict real log-concavity for the first complex 2-Hessian eigenfunction by verifying the Bian-Guan [5] structural condition. Their approaches are different from the constant rank theorem developed here.*

1.3. Strategy of the proof. Let us outline the fundamental framework for proving Theorem 1.1. We follow the approach of Caffarelli and Friedman [10], or Korevaar and Lewis [34], who established the power convexity of the torsion function and the first Dirichlet eigenfunction of the Laplacian by the microscopic convexity principle. It is well known that two key steps are required to implement this approach. The first is to obtain convexity estimates near the boundary, and the second is to establish the constant rank theorem. Recall the Dirichlet problem (1.3) for the complex Monge-Ampère equation considered here. Letting $v = -\sqrt{-u/2}$, we know that v satisfies the Dirichlet problem

$$\begin{cases} F(\nabla^2 v, \nabla v, v) = 0 & \text{in } \Omega, \\ v = 0 & \text{on } \partial\Omega, \end{cases}$$

where

$$\begin{aligned} F(\nabla^2 v, \nabla v, v) = & v^2 [(v_{11} + v_{33})(v_{22} + v_{44}) - (v_{12} + v_{34})(v_{21} + v_{43}) - (v_{14} - v_{32})(v_{41} - v_{23})] \\ & + v [(v_2^2 + v_4^2)(v_{11} + v_{33}) + (v_1^2 + v_3^2)(v_{22} + v_{44}) \\ & - (v_1 v_2 + v_3 v_4)(v_{12} + v_{34} + v_{21} + v_{43})] \end{aligned}$$

$$-(v_1v_4 - v_2v_3)(v_{14} - v_{32} + v_{41} - v_{23})] - 1.$$

For more details on the derivation of this equation, we refer to Section 2.1. Since the domain Ω is strictly convex, we can easily obtain the convexity estimates for v near the boundary $\partial\Omega$ (see Proposition 5.2 for a precise statement). It suffices to establish the constant rank theorem, namely that $\nabla^2 v$ has constant rank in Ω , for the transformed equation

$$(1.4) \quad F(\nabla^2 v, \nabla v, v) = 0.$$

Let us explain why establishing the constant rank theorem for equation (1.4) is particularly challenging. Note that equation (1.4) is non-homogeneous with respect to the variable $\nabla^2 v$. In general, it is difficult to verify the crucial inverse convexity structural condition (3.2) outlined in Bian and Guan [4] for non-homogeneous equations. Although Bryan, Ivaki and Scheuer [9] present a constant rank theorem for non-homogeneous equations, their conditions do not hold for k -Hessian type equations after a power transformation. Thus, we have to prove the constant rank theorem directly. This helps explain why, to date, convexity results obtained via this approach have been largely restricted to low dimensions [14, 26, 37, 39, 49].

In the complex setting, Hermitian matrices can be diagonalized by unitary transformations. Our goal, however, is to study the real convexity of solutions, for which the relevant object is the real Hessian $\nabla^2 v$. A major difficulty is that the unitary transformations naturally associated with the complex structure generally do not diagonalize $\nabla^2 v$, making direct computations difficult. To address this, we seek a coordinate system in which $\nabla^2 v$ is diagonal while both the equation and the auxiliary quantities are invariant under the coordinate transformation. By exploiting the Autonne-Takagi factorization, we construct a suitable complex linear transformation satisfying these requirements. The details are presented in Section 2.2.

Define

$$l = \min_{x \in \Omega} \text{rank}(\nabla^2 v(x)).$$

At a point $x_0 \in \Omega$ where $\nabla^2 v(x_0)$ attains the minimal rank l ($0 \leq l \leq 4$), we can choose suitable coordinates such that $\nabla^2 v(x_0)$ is diagonal. If $l = 0$ or $l = 1$, this would contradict equation (1.4), since the left-hand side of (1.4) is negative while the right-hand side is 0. If $l = 4$, it is evident that $\nabla^2 v(x)$ is of full rank for any $x \in \Omega$. Therefore, we only need to consider two cases, i.e.,

- $l = 2$ (Theorem 3.7);
- $l = 3$ (Theorem 4.1).

To establish the constant rank theorem, the key step is to choose an appropriate auxiliary function, derive suitable differential inequalities, and apply the strong maximum principle.

In their celebrated work, Bian and Guan [4] studied fully nonlinear elliptic equations

$$F(\nabla^2 v, \nabla v, v, x) = 0 \quad \text{in } \Omega \subset \mathbb{R}^n.$$

To establish the constant rank theorem, they introduced the auxiliary function

$$\phi(x) = \sigma_{l+1}(\nabla^2 v(x)) + q(\nabla^2 v(x)),$$

where

$$q(\nabla^2 v(x)) = \begin{cases} \frac{\sigma_{l+2}(\nabla^2 v(x))}{\sigma_{l+1}(\nabla^2 v(x))} & \text{if } \sigma_{l+1}(\nabla^2 v(x)) > 0, \\ 0 & \text{if } \sigma_{l+1}(\nabla^2 v(x)) = 0. \end{cases}$$

See Section 3.1 for further discussion of the elementary symmetric functions. For $l = 2$, this function involves $\sigma_3(\nabla^2 v)$. In our setting, the relevant changes of coordinates are complex linear transformations whose induced real transformations are used to diagonalize the real Hessian $\nabla^2 v$. Unlike unitary transformations, general complex linear transformations on $\mathbb{C}^2$ do not induce orthogonal transformations on $\mathbb{R}^4$ and therefore do not, in general, preserve the eigenvalues of $\nabla^2 v$ or the quantity $\sigma_3(\nabla^2 v)$. To overcome this difficulty, we introduce a 2×2 complex matrix K defined in (2.10), satisfying

$$\text{rank}(\nabla^2 v) = \text{rank}(K) + 2,$$

as stated in Lemma 2.1. This identity shows that $\nabla^2 v$ has constant rank if and only if K does. Under complex linear coordinate changes, K transforms by similarity. Hence its eigenvalues are preserved, and consequently $\text{tr}(K)$ and $\det(K)$ are invariant. This allows us to construct invariant auxiliary functions from $\text{tr}(K)$ and $\det(K)$, which play a central role in our proof. A detailed discussion is given in Section 2.2.

The case $l = 2$ corresponds to $\min_{x \in \Omega} \text{rank}(K(x)) = 0$. In the spirit of Bian and Guan [4], we define the auxiliary function

$$\phi(x) = \text{tr}(K(x)) + q(K(x)),$$

where

$$q(K(x)) = \begin{cases} \frac{\det(K(x))}{\text{tr}(K(x))} & \text{if } \text{tr}(K(x)) > 0, \\ 0 & \text{if } \text{tr}(K(x)) = 0. \end{cases}$$

As pointed out by Bian and Guan in [4], it is difficult to establish the maximum principle for general fully nonlinear elliptic equations using only the standard auxiliary function, which in our setting reduces to $\text{tr}(K)$. Even for the specific equation (1.4) considered here, choosing $\text{tr}(K)$ alone as the auxiliary function would lead to prohibitively complicated computations. The quotient component $q(K(x))$ is therefore essential to our argument.

The case $l = 3$ corresponds to $\min_{x \in \Omega} \text{rank}(K(x)) = 1$. This case is considerably more challenging. To prove the differential inequality for the auxiliary function $\det(K)$ in this case, we must verify the positive semidefiniteness of a quadratic form in six variables, most of whose coefficients are nonzero. This part is highly involved and technical. Our strategy is to perform a carefully chosen sequence of simultaneous row and column operations on

the corresponding 6×6 symmetric matrix. Fortunately, we can transform this 6×6 matrix into block diagonal form with two 3×3 blocks. By carefully computing and arranging the leading principal minors and applying a basic lemma (Lemma 2.2) from linear algebra, we finally complete the proof of Theorem 4.1.

The rest of this paper is organized as follows. In Section 2, we collect the necessary preliminaries, including the real and complex coordinate formulations, the notation used throughout the paper, the choice of suitable coordinates, and the construction of the auxiliary functions. In Sections 3 and 4, we establish the differential inequalities for the cases where the real Hessian $\nabla^2 v$ has minimal rank 2 and 3, respectively. These two sections constitute the main technical part of the paper. Finally, in Section 5, we first establish the constant rank theorem (Theorem 5.1) and then complete the proof of our main result, Theorem 1.1.

2. NOTATION AND PRELIMINARIES

In this section, the necessary preliminaries and notation for later use are provided. We first derive the real form of the two-dimensional complex Monge-Ampère equation under a power transformation. We then discuss the choice of adapted coordinates and the construction of invariant auxiliary functions. Finally, we recall a linear algebra lemma that will be used repeatedly.

2.1. Real form of the complex Monge-Ampère equation. This subsection is devoted to deriving the real form of the equation satisfied by v , which is obtained from the original complex Monge-Ampère equation through a power transformation.

We begin with the complex Monge-Ampère equation

$$(2.1) \quad \det(u_{i\bar{j}}) = 1 \quad \text{in } \Omega,$$

where $\Omega \subset \mathbb{C}^2$ is a bounded domain. To avoid confusion with the notation for the real Hessian introduced later, we shall write all derivatives in terms of the complex coordinates z_i and $\bar{z}_i$. In particular, we set

$$u_i = u_{z_i} = \frac{\partial u}{\partial z_i}, \quad u_{\bar{i}} = u_{\bar{z}_i} = \frac{\partial u}{\partial \bar{z}_i}.$$

Set $v = -\sqrt{-u/2}$. Direct computation gives

$$u_{z_i} = -4vv_{z_i}, \quad u_{z_i\bar{z}_j} = -4(vv_{z_i\bar{z}_j} + v_{z_i}v_{\bar{z}_j}),$$

so that the equation (2.1) becomes

$$(2.2) \quad \det(vv_{z_i\bar{z}_j} + v_{z_i}v_{\bar{z}_j}) = \frac{1}{16}.$$

To work with real coordinates, we express the complex derivatives in terms of $z_i = x_i + \sqrt{-1}y_i$. For the real-valued function v defined on Ω , the complex derivatives v_{z_i} and

$v_{\bar{z}_i}$ are given by

$$(2.3) \quad v_{z_i} = \frac{\partial v}{\partial z_i} = \frac{1}{2} \left(\frac{\partial v}{\partial x_i} - \sqrt{-1} \frac{\partial v}{\partial y_i} \right) = \frac{1}{2} (v_{x_i} - \sqrt{-1} v_{y_i})$$

and

$$(2.4) \quad v_{\bar{z}_i} = \frac{\partial v}{\partial \bar{z}_i} = \frac{1}{2} \left(\frac{\partial v}{\partial x_i} + \sqrt{-1} \frac{\partial v}{\partial y_i} \right) = \frac{1}{2} (v_{x_i} + \sqrt{-1} v_{y_i}).$$

Furthermore, we have

$$(2.5) \quad v_{z_i \bar{z}_j} = \frac{\partial^2 v}{\partial z_i \partial \bar{z}_j} = \frac{1}{4} (v_{x_i x_j} + v_{y_i y_j}) + \frac{1}{4} \sqrt{-1} (v_{x_i y_j} - v_{y_i x_j})$$

and

$$v_{z_i z_j} = \frac{\partial^2 v}{\partial z_i \partial z_j} = \frac{1}{4} (v_{x_i x_j} - v_{y_i y_j}) - \frac{1}{4} \sqrt{-1} (v_{x_i y_j} + v_{y_i x_j}).$$

Let $A = (v_{z_i \bar{z}_j})$, $B = (v_{z_i z_j})$, and write the real Hessian of v in block form as

$$\nabla^2 v = \begin{pmatrix} U & V \\ V^T & W \end{pmatrix},$$

where $U = (v_{x_i x_j})$, $V = (v_{x_i y_j})$, and $W = (v_{y_i y_j})$. Then

$$A = \frac{1}{4} (U + W) + \frac{\sqrt{-1}}{4} (V - V^T),$$

and

$$B = \frac{1}{4} (U - W) - \frac{\sqrt{-1}}{4} (V + V^T).$$

Since u is strictly plurisubharmonic and $v < 0$, it follows from $u_{z_i \bar{z}_j} = -4(v v_{z_i \bar{z}_j} + v_{z_i} v_{\bar{z}_j})$ that $A = (v_{z_i \bar{z}_j})$ is positive definite. In fact, for any nonzero vector $\xi \in \mathbb{C}^2$,

$$\sum_{i,j} v_{z_i \bar{z}_j} \xi_i \bar{\xi}_j = -\frac{1}{4v} \sum_{i,j} u_{z_i \bar{z}_j} \xi_i \bar{\xi}_j - \frac{1}{v} \left| \sum_i v_{z_i} \xi_i \right|^2 > 0.$$

In particular, A is invertible.

For the sake of convenience, we adopt the following notation for real partial derivatives, which will be used throughout the paper:

$$\begin{aligned} v_{x_1} &= v_1, \quad v_{x_2} = v_2, \quad v_{y_1} = v_3, \quad v_{y_2} = v_4, \\ v_{x_1 x_1} &= v_{11}, \quad v_{y_1 y_1} = v_{33}, \quad v_{x_1 y_1} = v_{13}, \quad \dots \end{aligned}$$

Substituting (2.5) into (2.2), we obtain the following real equation

$$(2.6) \quad F(\nabla^2 v, \nabla v, v) = 0,$$

where

$$F(\nabla^2 v, \nabla v, v) = v^2 [(v_{11} + v_{33})(v_{22} + v_{44}) - (v_{12} + v_{34})(v_{21} + v_{43}) - (v_{14} - v_{32})(v_{41} - v_{23})]$$

$$\begin{aligned}
& + v [(v_2^2 + v_4^2)(v_{11} + v_{33}) + (v_1^2 + v_3^2)(v_{22} + v_{44}) \\
& - (v_1v_2 + v_3v_4)(v_{12} + v_{34} + v_{21} + v_{43}) \\
& - (v_1v_4 - v_2v_3)(v_{14} - v_{32} + v_{41} - v_{23})] - 1.
\end{aligned}$$

Hence v satisfies the Dirichlet problem

$$(2.7) \quad \begin{cases} F(\nabla^2 v, \nabla v, v) = 0 & \text{in } \Omega, \\ v = 0 & \text{on } \partial\Omega. \end{cases}$$

For simplicity, we denote

$$\begin{aligned}
F^{ij} &= \frac{\partial F}{\partial v_{ij}}, & F^{v_k} &= \frac{\partial F}{\partial v_k}, & F^v &= \frac{\partial F}{\partial v}, \\
F^{ij,kl} &= \frac{\partial^2 F}{\partial v_{ij} \partial v_{kl}}, & F^{ij,v_k} &= \frac{\partial^2 F}{\partial v_{ij} \partial v_k}, & F^{ij,v} &= \frac{\partial^2 F}{\partial v_{ij} \partial v}, \\
F^{v_k, v_l} &= \frac{\partial^2 F}{\partial v_k \partial v_l}, & F^{v_k, v} &= \frac{\partial^2 F}{\partial v_k \partial v}, & F^{v, v} &= \frac{\partial^2 F}{\partial v^2}, \quad \text{etc.}
\end{aligned}$$

The above notation will be used throughout Sections 3 and 4.

We end this subsection by verifying the ellipticity of equation (2.6). Direct differentiation gives

$$(2.8) \quad (F^{ij}) = \begin{pmatrix} F^{11} & F^{12} & 0 & -F^{23} \\ F^{12} & F^{22} & F^{23} & 0 \\ 0 & F^{23} & F^{11} & F^{12} \\ -F^{23} & 0 & F^{12} & F^{22} \end{pmatrix},$$

where

$$\begin{aligned}
F^{11} &= v^2(v_{22} + v_{44}) + v(v_2^2 + v_4^2), \\
F^{22} &= v^2(v_{11} + v_{33}) + v(v_1^2 + v_3^2), \\
F^{12} &= -v^2(v_{21} + v_{43}) - v(v_1v_2 + v_3v_4), \\
F^{23} &= v^2(v_{14} - v_{32}) + v(v_1v_4 - v_2v_3).
\end{aligned}$$

Since the solution u of equation (2.1) is strictly plurisubharmonic, it follows that $u_{z_1\bar{z}_1}$ and $u_{z_2\bar{z}_2}$ are positive. Then, by the identity $u_{z_i\bar{z}_j} = -4(vv_{z_i\bar{z}_j} + v_{z_i}v_{\bar{z}_j})$ and (2.3)-(2.5), we obtain

$$v(v_{11} + v_{33}) + v_1^2 + v_3^2 < 0$$

and

$$v(v_{22} + v_{44}) + v_2^2 + v_4^2 < 0.$$

Together with $v < 0$, these imply that $F^{11} > 0$ and $F^{22} > 0$. Moreover, a direct calculation using equation (2.6) yields

$$(2.9) \quad F^{11}F^{22} - (F^{12})^2 - (F^{23})^2 = v^2.$$

Denote the four leading principal minors of (F^{ij}) by P_1, P_2, P_3 and P_4 , respectively. Then we have

$$\begin{aligned} P_1 &= F^{11} > 0, \\ P_2 &= F^{11}F^{22} - (F^{12})^2 = v^2 + (F^{23})^2 > 0, \\ P_3 &= F^{11}(F^{11}F^{22} - (F^{12})^2 - (F^{23})^2) = v^2F^{11} > 0, \end{aligned}$$

and

$$P_4 = (F^{11}F^{22} - (F^{12})^2 - (F^{23})^2)^2 = v^4 > 0.$$

Thus, the matrix (F^{ij}) is positive definite, and equation (2.6) is elliptic with respect to the real Hessian variables.

2.2. Choice of the auxiliary functions. The goal of this subsection is to construct suitable invariant auxiliary functions for the proof of the constant rank theorem for the real Hessian $\nabla^2 v$. The construction relies on two fundamental facts, namely the existence of an adapted coordinate system in which $\nabla^2 v$ can be diagonalized at a given point, and the fact that the transformed complex Monge-Ampère equation is invariant under such coordinate transformations.

We define the auxiliary matrix K by

$$(2.10) \quad K = I - B\overline{A}^{-1}\overline{B}A^{-1},$$

where I denotes the identity matrix of size 2×2 , $A = (v_{z_i \bar{z}_j})_{2 \times 2}$ and $B = (v_{z_i z_j})_{2 \times 2}$ (see Subsection 2.1). We first establish the following rank identity between the real Hessian $\nabla^2 v$ and the auxiliary matrix K .

Lemma 2.1. *The ranks of the real Hessian $\nabla^2 v$ and the auxiliary matrix K are related by*

$$\text{rank}(\nabla^2 v) = \text{rank}(K) + 2.$$

Proof. The standard real-complex correspondence gives the following matrix identity

$$(2.11) \quad 4 \begin{pmatrix} A & B \\ \overline{B} & \overline{A} \end{pmatrix} = \begin{pmatrix} I & -\sqrt{-1}I \\ I & \sqrt{-1}I \end{pmatrix} \begin{pmatrix} U & V \\ V^T & W \end{pmatrix} \begin{pmatrix} I & I \\ \sqrt{-1}I & -\sqrt{-1}I \end{pmatrix} \triangleq T \nabla^2 v T^H,$$

where

$$T = \begin{pmatrix} I & -\sqrt{-1}I \\ I & \sqrt{-1}I \end{pmatrix}$$

is an invertible complex matrix and T^H is the conjugate transpose of T . Thus

$$\text{rank}(\nabla^2 v) = \text{rank} \begin{pmatrix} A & B \\ \overline{B} & \overline{A} \end{pmatrix}.$$

On the other hand, a standard block decomposition yields

$$(2.12) \quad \begin{pmatrix} I & -B\overline{A^{-1}} \\ 0 & I \end{pmatrix} \begin{pmatrix} A & B \\ \overline{B} & \overline{A} \end{pmatrix} \begin{pmatrix} I & 0 \\ -\overline{A^{-1}}\overline{B} & I \end{pmatrix} = \begin{pmatrix} A - B\overline{A^{-1}}\overline{B} & 0 \\ 0 & \overline{A} \end{pmatrix},$$

which implies the rank identity

$$\text{rank} \begin{pmatrix} A & B \\ \overline{B} & \overline{A} \end{pmatrix} = \text{rank}(A - B\overline{A^{-1}}\overline{B}) + \text{rank}(A).$$

Since A is an invertible 2×2 matrix and K is defined by (2.10), we obtain

$$\text{rank}(\nabla^2 v) = \text{rank}(K) + 2.$$

□

To simplify the local computations, we choose coordinates in which the real Hessian $\nabla^2 v$ is diagonal at a given point $x_0 \in \Omega$. According to Lemma A.4 in Hu [24], which is based on the Autonne-Takagi factorization (see also Lemma A.3 therein), there exists an invertible matrix P such that

$$PAP^H = I, \quad PBP^T = \Lambda,$$

where Λ is a diagonal matrix with nonnegative diagonal entries. We now recall the explicit expressions of the complex Hessian blocks A and B in terms of the real second derivatives of v , namely

$$(2.13) \quad A = \begin{pmatrix} \frac{1}{4}(v_{11} + v_{33}) & \frac{1}{4}(v_{12} + v_{34}) + \frac{\sqrt{-1}}{4}(v_{14} - v_{32}) \\ \frac{1}{4}(v_{21} + v_{43}) + \frac{\sqrt{-1}}{4}(v_{23} - v_{41}) & \frac{1}{4}(v_{22} + v_{44}) \end{pmatrix}$$

and

$$(2.14) \quad B = \begin{pmatrix} \frac{1}{4}(v_{11} - v_{33}) - \frac{\sqrt{-1}}{4}(v_{13} + v_{31}) & \frac{1}{4}(v_{12} - v_{34}) - \frac{\sqrt{-1}}{4}(v_{14} + v_{32}) \\ \frac{1}{4}(v_{21} - v_{43}) - \frac{\sqrt{-1}}{4}(v_{23} + v_{41}) & \frac{1}{4}(v_{22} - v_{44}) - \frac{\sqrt{-1}}{4}(v_{24} + v_{42}) \end{pmatrix}.$$

The relation $PAP^H = I$ implies

$$v_{12} + v_{34} = 0, \quad v_{14} - v_{32} = 0,$$

while the relation $PBP^T = \Lambda$ gives

$$v_{12} - v_{34} = 0, \quad v_{14} + v_{32} = 0.$$

Combining these relations and using the symmetry of the real Hessian, we have

$$v_{12} = v_{34} = v_{14} = v_{23} = 0.$$

Moreover, since $PBP^T = \Lambda$ is a real diagonal matrix, we also have

$$v_{13} = v_{24} = 0.$$

This confirms the existence of an adapted coordinate system at x_0 under which the real Hessian $\nabla^2 v(x_0)$ is diagonal.

We further normalize the coordinate transformation P by requiring $|\det(P)| = 1$. Under this normalization, we have

$$PAP^H = \sqrt{\det(A)}I, \quad PBP^T = \sqrt{\det(A)}\Lambda,$$

and

$$\det(P(vv_{z_i\bar{z}_j} + v_{z_i}v_{\bar{z}_j})P^H) = \det(vv_{z_i\bar{z}_j} + v_{z_i}v_{\bar{z}_j}) = \frac{1}{16}.$$

Thus, equation (2.2) is invariant under complex linear transformations P satisfying $|\det(P)| = 1$.

Applying the above transformation, the matrices A and B transform as

$$A \mapsto PAP^H, \quad B \mapsto PBP^T.$$

Substituting these into the formula of K , we obtain

$$\begin{aligned} \tilde{K} &= I - (PBP^T)(\overline{PAP^H})^{-1}(PBP^T)(PAP^H)^{-1} \\ &= I - PBP^T(\overline{P^H})^{-1}A^{-1}P^{-1}PBP^T(P^H)^{-1}A^{-1}P^{-1} \\ &= I - P\overline{BA^{-1}}\overline{BA}^{-1}P^{-1} \\ &= P(I - \overline{BA^{-1}}\overline{BA}^{-1})P^{-1} \\ &= PKP^{-1}. \end{aligned}$$

Hence K transforms by a similarity transformation. Consequently, its eigenvalues are preserved, and therefore both $\text{tr}(K)$ and $\det(K)$ are invariant under this transformation.

Moreover, the real Hessian $\nabla^2 v$ is positive definite (respectively, positive semidefinite) if and only if all the eigenvalues of K are positive (respectively, nonnegative); see [24, Lemma A.6].

As discussed in Section 1.3, it suffices to prove the constant rank theorem for $\nabla^2 v$ when its minimal rank is 2 or 3. In view of Lemma 2.1, the constant rank problem for $\nabla^2 v$ is reduced to that for the auxiliary matrix K , namely proving the constant rank theorem in the cases $\min_{x \in \Omega} \text{rank}(K(x)) = 0$ and $\min_{x \in \Omega} \text{rank}(K(x)) = 1$, respectively. We therefore select suitable auxiliary functions for these two cases.

• $\min_{x \in \Omega} \text{rank}(K(x)) = 0$. In this case, in the spirit of Bian and Guan [4], we employ the quotient-type auxiliary function

$$(2.15) \quad \phi(x) = \text{tr}(K(x)) + q(K(x)),$$

where

$$q(K(x)) = \begin{cases} \frac{\det(K(x))}{\text{tr}(K(x))} & \text{if } \text{tr}(K(x)) > 0, \\ 0 & \text{if } \text{tr}(K(x)) = 0. \end{cases}$$

The invariance of $\text{tr}(K(x))$ and $\det(K(x))$ implies the invariance of the auxiliary function ϕ . Without loss of generality, ϕ can be computed in the above diagonalized coordinate system.

- $\min_{x \in \Omega} \text{rank}(K(x)) = 1$. We use the standard auxiliary function

$$(2.16) \quad \phi = \det(K(x)),$$

as in Caffarelli and Friedman [10], Korevaar and Lewis [34], and other related works. In fact, this is also consistent with the construction of Bian and Guan [4].

We shall apply these two auxiliary functions in Sections 3 and 4 to obtain the differential inequalities stated in Theorems 3.7 and 4.1, which allow us to use the strong maximum principle to establish the constant rank theorem.

2.3. A linear algebra lemma. We conclude this section with a useful lemma from linear algebra. Although this lemma is standard and can be found in many linear algebra textbooks, it plays an important role in the proof of the constant rank theorem in the cases of minimal rank 2 and 3. For completeness, we include a proof here.

Lemma 2.2. *Let M be an $n \times n$ real symmetric matrix. Assume that the first $n - 1$ leading principal minors of M are nonzero, and denote them by $P_1, \dots, P_{n-1}$. Then there exists an $n \times n$ invertible matrix C such that*

$$C^T M C = \text{diag} \left(P_1, \frac{P_2}{P_1}, \dots, \frac{P_n}{P_{n-1}} \right),$$

where $P_n = \det(M)$.

Proof. We proceed by induction on n . The result is evident when $n = 1$. Assume that the statement holds for $n - 1$, and let

$$M = \begin{pmatrix} M_{n-1} & \alpha \\ \alpha^T & a_{nn} \end{pmatrix},$$

where M_{n-1} is the $(n - 1) \times (n - 1)$ leading principal submatrix of M . Since $\det(M_{n-1}) = P_{n-1} \neq 0$, we may define an $n \times n$ invertible matrix

$$C_1 = \begin{pmatrix} I_{n-1} & -M_{n-1}^{-1}\alpha \\ O & 1 \end{pmatrix}.$$

Then we have

$$C_1^T M C_1 = \begin{pmatrix} M_{n-1} & O \\ O & a_{nn} - \alpha^T M_{n-1}^{-1} \alpha \end{pmatrix}.$$

Note that $\det(C_1) = 1$. It follows that

$$\det(M) = (a_{nn} - \alpha^T M_{n-1}^{-1} \alpha) \det(M_{n-1}),$$

that is,

$$a_{nn} - \alpha^T M_{n-1}^{-1} \alpha = \frac{P_n}{P_{n-1}}.$$

By the inductive hypothesis, there exists an $(n-1) \times (n-1)$ invertible matrix C_2 such that

$$C_2^T M_{n-1} C_2 = \text{diag} \left(P_1, \frac{P_2}{P_1}, \dots, \frac{P_{n-1}}{P_{n-2}} \right).$$

If we let $C_3 = \begin{pmatrix} C_2 & O \\ O & 1 \end{pmatrix}$ and $C = C_1 C_3$, then we have

$$C^T M C = \text{diag} \left(P_1, \frac{P_2}{P_1}, \dots, \frac{P_n}{P_{n-1}} \right).$$

□

At this stage, we have completed the preparation of the necessary notation, the choice of adapted coordinates, and the construction of the auxiliary functions that will be used in the sequel.

3. DIFFERENTIAL INEQUALITY: THE CASE OF MINIMAL RANK 2

This section is devoted to the proof of the constant rank theorem for the real Hessian $\nabla^2 v$ in the case where it attains its minimal rank 2, or equivalently, where the auxiliary matrix K has minimal rank 0. More precisely, our argument is inspired by the microscopic convexity principle of Bian and Guan [4]. In contrast to their construction based on the elementary symmetric functions of the Hessian, we introduce a new quotient-type auxiliary function adapted to the structure of the matrix K , as defined in (2.15). The first step is to perform a careful computation of the first and second order derivatives of the quotient term in this auxiliary function. Based on these expansions, we establish a strong maximum principle for ϕ , which ultimately yields the constant rank theorem in the present setting.

Throughout this section and the following one, we adopt the convention that all summation indices i, j, k, l, p, q, r, s and $\alpha, \beta, \gamma, \delta$ range from 1 to 4, unless otherwise indicated.

3.1. Estimates for the quotient component. In this subsection, we focus on the quotient term in the auxiliary function (2.15), which requires a careful computation.

We first recall the definition of the k -th elementary symmetric functions. For $1 \leq k \leq n$ and $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n$, we define

$$\sigma_k(\lambda) = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_k}.$$

We also adopt the convention that $\sigma_0(\lambda) = 1$ and $\sigma_{-1}(\lambda) = \sigma_{n+1}(\lambda) = 0$. The definition of σ_k for vectors in $\mathbb{R}^n$ can be extended to $n \times n$ symmetric matrices in a natural way by letting

$\sigma_k(W) = \sigma_k(\lambda(W))$, where $\lambda(W) = (\lambda_1(W), \lambda_2(W), \dots, \lambda_n(W))$ are the eigenvalues of the symmetric matrix W . We denote by $\sigma_{k-1}(\lambda|i)$ the $(k-1)$ -th elementary symmetric function obtained by setting $\lambda_i = 0$, and by $\sigma_{k-2}(\lambda|ij)$ ($i \neq j$) the $(k-2)$ -th elementary symmetric function obtained by setting $\lambda_i = \lambda_j = 0$. When W is diagonal, we use the same notation $\sigma_{k-1}(W|i)$ and $\sigma_{k-2}(W|ij)$ for the corresponding symmetric functions of its diagonal entries.

The first and second derivatives of the elementary symmetric functions are given in the following lemma.

Lemma 3.1. *Suppose that $W = (W_{ij})_{n \times n}$ is diagonal. For $k = 1, 2, \dots, n$, we have*

$$\frac{\partial \sigma_k(W)}{\partial W_{ij}} = \begin{cases} \sigma_{k-1}(W|i) & \text{if } i = j \\ 0 & \text{if } i \neq j, \end{cases}$$

and

$$\frac{\partial^2 \sigma_k(W)}{\partial W_{ij} \partial W_{pq}} = \begin{cases} \sigma_{k-2}(W|ip) & \text{if } i = j, p = q, i \neq p, \\ -\sigma_{k-2}(W|ip) & \text{if } i = q, j = p, i \neq p, \\ 0, & \text{otherwise.} \end{cases}$$

We briefly review the constant rank theorem of Bian and Guan [4], which has been mentioned several times in the preceding discussion. They introduced the quotient-type auxiliary function

$$(3.1) \quad \phi(x) = \sigma_{l+1}(\nabla^2 v(x)) + q(\nabla^2 v(x)),$$

where

$$q(\nabla^2 v(x)) = \begin{cases} \frac{\sigma_{l+2}(\nabla^2 v(x))}{\sigma_{l+1}(\nabla^2 v(x))} & \text{if } \sigma_{l+1}(\nabla^2 v(x)) > 0, \\ 0 & \text{if } \sigma_{l+1}(\nabla^2 v(x)) = 0. \end{cases}$$

Using the auxiliary function (3.1), they established the constant rank theorem for convex solutions of the general fully nonlinear elliptic equation

$$F(\nabla^2 v, \nabla v, v, x) = 0 \quad \text{in } \Omega \subset \mathbb{R}^n,$$

under the structural condition that

$$(3.2) \quad F(M^{-1}, p, v, x) \text{ is locally convex in } (M, v, x) \text{ for each } p.$$

Below we recall the notation from Bian and Guan [4] and adapt it to the present setting. Suppose that $\nabla^2 v(x)$ attains its minimal rank 2 at a point $x_0 \in \Omega$. Let $\mathcal{O}$ be an open neighborhood of x_0 . For each $x \in \mathcal{O}$, let $\lambda(x) = (\lambda_1(x), \lambda_2(x), \lambda_3(x), \lambda_4(x))$ be the eigenvalues of $\nabla^2 v(x)$. Then there exists a positive constant c independent of x , such that $\lambda_1(x) \geq \lambda_2(x) \geq c$. Let $G = \{1, 2\}$ and $B = \{3, 4\}$ be the sets of “good” and “bad” indices, respectively. Whenever no confusion is likely to arise, we also use G and B to denote the corresponding sets of “good” and “bad” eigenvalues of $\nabla^2 v(x)$. Note that, for any $\delta > 0$, we may choose $\mathcal{O}$ sufficiently small such that $\lambda_i(x) < \delta$ for all $i \in B$ and $x \in \mathcal{O}$.

Finally, for two functions $h(x)$ and $k(x)$ defined on $\mathcal{O}$, the notation $h(x) = O(k(x))$ means that there exists a positive constant C such that $|h(x)| \leq C|k(x)|$.

By Lemma 2.1, we have the rank identity

$$\text{rank}(\nabla^2 v) = \text{rank}(K) + 2.$$

Therefore, the case where $\nabla^2 v$ attains its minimal rank 2 is equivalent to the case where the auxiliary matrix K attains its minimal rank 0. Motivated by the discussions in Section 1.3 and Section 2.2, we now apply the Bian–Guan framework to the auxiliary matrix K , rather than to the real Hessian $\nabla^2 v$. Accordingly, we introduce the auxiliary function

$$(3.3) \quad \phi(x) = \text{tr}(K(x)) + q(K(x)),$$

where

$$q(K(x)) = \begin{cases} \frac{\det(K(x))}{\text{tr}(K(x))} & \text{if } \text{tr}(K(x)) > 0, \\ 0 & \text{if } \text{tr}(K(x)) = 0. \end{cases}$$

To facilitate the subsequent computations, we rewrite the auxiliary function ϕ in terms of the entries of $\nabla^2 v$. According to the definition of the matrix K in (2.10) and identities (2.11)–(2.12), we have

$$(3.4) \quad \det(K) = \frac{1}{16}(\det(A))^{-2} \det(\nabla^2 v).$$

Furthermore, we may write

$$K = (\det(A))^{-2}((\det(A))^2 I - B\overline{A^*}BA^*),$$

where A^* denotes the adjugate matrix of A . Consequently,

$$\text{tr}(K) = (\det(A))^{-2} \text{tr}((\det(A))^2 I - B\overline{A^*}BA^*).$$

Define $\psi = \text{tr}((\det(A))^2 I - B\overline{A^*}BA^*)$. Then the auxiliary function ϕ in (3.3) can be written as

$$(3.5) \quad \phi = (\det(A))^{-2} \psi + \frac{\det(\nabla^2 v)}{16\psi}.$$

The introduction of ψ removes the denominator $(\det(A))^{-2}$ from $\text{tr}(K)$, making the subsequent differentiation considerably simpler. For notational convenience, we continue to denote

$$(3.6) \quad q(\nabla^2 v) = \frac{\det(\nabla^2 v)}{16\psi}.$$

Using the explicit expressions of the matrices A and B in (2.13) and (2.14), a straightforward calculation yields

$$B\overline{A^*} = \frac{1}{16} \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where

$$a = (v_{11} - v_{33})(v_{22} + v_{44}) - (v_{12} - v_{34})(v_{21} + v_{43}) + (v_{14} + v_{32})(v_{23} - v_{41})$$

$$\begin{aligned}
& -\sqrt{-1}(v_{13}+v_{31})(v_{22}+v_{44})+\sqrt{-1}(v_{14}+v_{32})(v_{21}+v_{43})+\sqrt{-1}(v_{12}-v_{34})(v_{23}-v_{41}), \\
b &= (v_{12}-v_{34})(v_{11}+v_{33})-(v_{11}-v_{33})(v_{12}+v_{34})+(v_{13}+v_{31})(v_{14}-v_{32}) \\
& -\sqrt{-1}(v_{14}+v_{32})(v_{11}+v_{33})+\sqrt{-1}(v_{13}+v_{31})(v_{12}+v_{34})+\sqrt{-1}(v_{11}-v_{33})(v_{14}-v_{32}), \\
c &= (v_{21}-v_{43})(v_{22}+v_{44})-(v_{22}-v_{44})(v_{21}+v_{43})+(v_{24}+v_{42})(v_{23}-v_{41}) \\
& -\sqrt{-1}(v_{23}+v_{41})(v_{22}+v_{44})+\sqrt{-1}(v_{24}+v_{42})(v_{21}+v_{43})+\sqrt{-1}(v_{22}-v_{44})(v_{23}-v_{41}), \\
d &= (v_{22}-v_{44})(v_{11}+v_{33})-(v_{21}-v_{43})(v_{12}+v_{34})+(v_{23}+v_{41})(v_{14}-v_{32}) \\
& -\sqrt{-1}(v_{24}+v_{42})(v_{11}+v_{33})+\sqrt{-1}(v_{23}+v_{41})(v_{12}+v_{34})+\sqrt{-1}(v_{21}-v_{43})(v_{14}-v_{32}).
\end{aligned}$$

Since $\overline{BA^*} = \overline{BA^*}$, it follows that

$$\mathrm{tr}(\overline{BA^*}BA^*) = \frac{1}{16^2}(|a|^2 + |d|^2 + 2\Re(b\bar{c})).$$

On the other hand, we have

$$\det(A) = \frac{1}{16}((v_{11}+v_{33})(v_{22}+v_{44})-(v_{12}+v_{34})(v_{21}+v_{43})-(v_{14}-v_{32})(v_{41}-v_{23})).$$

Combining these identities with the definition of ψ , we obtain

$$\begin{aligned}
(3.7) \quad \psi &= \mathrm{tr}((\det(A))^2 I - \overline{BA^*}BA^*) \\
&= \frac{2}{16^2}((v_{11}+v_{33})(v_{22}+v_{44})-(v_{12}+v_{34})(v_{21}+v_{43})-(v_{14}-v_{32})(v_{41}-v_{23}))^2 \\
&\quad - \frac{1}{16^2} \left[((v_{11}-v_{33})(v_{22}+v_{44})-(v_{12}-v_{34})(v_{21}+v_{43})+(v_{14}+v_{32})(v_{23}-v_{41}))^2 \right. \\
&\quad + ((v_{22}-v_{44})(v_{11}+v_{33})-(v_{21}-v_{43})(v_{12}+v_{34})+(v_{23}+v_{41})(v_{14}-v_{32}))^2 \\
&\quad + (- (v_{13}+v_{31})(v_{22}+v_{44})+(v_{14}+v_{32})(v_{21}+v_{43})+(v_{12}-v_{34})(v_{23}-v_{41}))^2 \\
&\quad + (- (v_{24}+v_{42})(v_{11}+v_{33})+(v_{23}+v_{41})(v_{12}+v_{34})+(v_{21}-v_{43})(v_{14}-v_{32}))^2 \\
&\quad + 2((v_{12}-v_{34})(v_{11}+v_{33})-(v_{11}-v_{33})(v_{12}+v_{34})+(v_{13}+v_{31})(v_{14}-v_{32})) \\
&\quad \cdot ((v_{21}-v_{43})(v_{22}+v_{44})-(v_{22}-v_{44})(v_{21}+v_{43})+(v_{24}+v_{42})(v_{23}-v_{41})) \\
&\quad + 2(- (v_{14}+v_{32})(v_{11}+v_{33})+(v_{13}+v_{31})(v_{12}+v_{34})+(v_{11}-v_{33})(v_{14}-v_{32})) \\
&\quad \cdot (- (v_{23}+v_{41})(v_{22}+v_{44})+(v_{24}+v_{42})(v_{21}+v_{43})+(v_{22}-v_{44})(v_{23}-v_{41})) \left. \right].
\end{aligned}$$

To ensure that ψ does not vanish in (3.6), we follow the perturbation argument of Bian and Guan [4]. For a sufficiently small $\varepsilon > 0$, define $v_\varepsilon(x) = v(x) + \frac{\varepsilon}{2}|x|^2$. Then $\nabla^2 v_\varepsilon = \nabla^2 v + \varepsilon I_4$, where I_4 is the 4×4 identity matrix. Let A_ε and B_ε be the perturbed quantities obtained by replacing $\nabla^2 v$ with $\nabla^2 v_\varepsilon$ in the definitions (2.13) and (2.14). Accordingly, we

define

$$\phi_\varepsilon = (\det(A_\varepsilon))^{-2}\psi_\varepsilon + q_\varepsilon(\nabla^2 v_\varepsilon),$$

where

$$\psi_\varepsilon = \operatorname{tr}((\det(A_\varepsilon))^2 I - B_\varepsilon \overline{A_\varepsilon^*} B_\varepsilon A_\varepsilon^*) \quad \text{and} \quad q_\varepsilon(\nabla^2 v_\varepsilon) = \frac{\sigma_4(\nabla^2 v_\varepsilon)}{16\psi_\varepsilon}.$$

We will work with the perturbed auxiliary function ϕ_ε to derive uniform C^2 estimates independent of ε . Alternatively, one may work directly with ϕ at points where $\psi \neq 0$, since all constants obtained below are independent of the chosen point. For simplicity of notation, we will write v, A, B, K, ψ, ϕ in place of $v_\varepsilon, A_\varepsilon, B_\varepsilon, K_\varepsilon, \psi_\varepsilon, \phi_\varepsilon$, respectively. Unless otherwise specified, all estimates below are uniform in ε .

Without loss of generality, we work at an arbitrary fixed point $x \in \mathcal{O}$ in the coordinate system introduced in Section 2.2, in which $\nabla^2 v$ is diagonal and satisfies

$$(3.8) \quad v_{11} + v_{33} = v_{22} + v_{44}.$$

In this coordinate system, the formula (3.7) reduces to

$$(3.9) \quad \psi = \frac{4}{16^2}(v_{22} + v_{44})^2 v_{11} v_{33} + \frac{4}{16^2}(v_{11} + v_{33})^2 v_{22} v_{44}.$$

Choosing $\mathcal{O}$ sufficiently small, we may assume that

$$C \geq v_{11} \geq v_{22} \geq c$$

throughout $\mathcal{O}$, where the positive constants C and c are independent of x . Consequently,

$$(3.10) \quad \psi \geq \frac{4}{16^2}(v_{22}^2 v_{11} v_{33} + v_{11}^2 v_{22} v_{44}) \geq c(v_{33} + v_{44}).$$

Moreover, since the perturbation guarantees that $v_{33}, v_{44} \geq \varepsilon$, we further obtain

$$(3.11) \quad \psi(x) \geq c\varepsilon \quad \text{for all } x \in \mathcal{O},$$

where c is independent of ε .

We next compare ϕ with $v_{33} + v_{44}$. By (3.9), one has

$$\phi \geq (\det(A))^{-2}\psi = 4(v_{11} + v_{33})^{-2}v_{11}v_{33} + 4(v_{22} + v_{44})^{-2}v_{22}v_{44}.$$

Since v_{11} and v_{22} are uniformly bounded from above and below, it follows that

$$\phi \geq c(v_{33} + v_{44}).$$

As immediate consequences, we have

$$(3.12) \quad v_{33} = O(\phi), \quad v_{44} = O(\phi).$$

By (3.8), we also conclude that

$$(3.13) \quad v_{22} = v_{11} + O(\phi).$$

On the other hand, it is clear that

$$\psi \leq C(v_{33} + v_{44}).$$

Combining this estimate with (3.10), we have

$$\phi = (\det A)^{-2} \psi + \frac{\sigma_4(\nabla^2 v)}{16\psi} \leq C'(v_{33} + v_{44}) + C'' \frac{v_{33}v_{44}}{v_{33} + v_{44}} \leq C(v_{33} + v_{44}).$$

Thus, we obtain

$$(3.14) \quad c\phi \leq v_{33} + v_{44} \leq C\phi.$$

To prepare for the estimates of the quotient component, we first determine the leading-order behavior of ψ and the asymptotic expansions of its inverse powers. By (3.9), together with the estimates (3.12) and (3.13), we derive

$$(3.15) \quad \psi = \frac{4}{16^2} v_{11}^3 (v_{33} + v_{44}) + O(\phi^2).$$

Denote $\psi_0 = \frac{4}{16^2} v_{11}^3 (v_{33} + v_{44})$. Then we have $\psi_0 = O(\phi)$ and $\psi - \psi_0 = O(\phi^2)$. By (3.14), we also have $\frac{\psi - \psi_0}{\psi_0} = O(\phi)$. Applying the Taylor expansion of $(1 + t)^{-m}$ ($m = 1, 2, 3$) at $t = 0$, we obtain

$$(3.16) \quad \begin{aligned} \psi^{-m} &= \psi_0^{-m} \left(1 + \frac{\psi - \psi_0}{\psi_0} \right)^{-m} \\ &= \psi_0^{-m} + O(\phi^{-m+1}). \end{aligned}$$

We are now ready to compute the first-order derivatives of q . Throughout this section, we use the convention that the index $\alpha = 1, 2, 3, 4$ corresponds to the variables x_1, x_2, y_1, y_2 , respectively.

Lemma 3.2. *Let $q(\nabla^2 v)$ be defined as in (3.6). Assume that $\nabla^2 v$ is diagonal at the point under consideration. Then*

$$q_\alpha = 4v_{11}^{-1} \frac{v_{44}^2}{\sigma_1^2(B)} v_{33\alpha} + 4v_{11}^{-1} \frac{v_{33}^2}{\sigma_1^2(B)} v_{44\alpha} + O(\phi).$$

Proof. By the definition of q , a direct computation gives

$$q_\alpha = -\frac{1}{16} \psi^{-2} \psi_\alpha \det(\nabla^2 v) + \frac{1}{16} \psi^{-1} \sum_{i,j} \frac{\partial \det(\nabla^2 v)}{\partial v_{ij}} v_{ij\alpha}.$$

Since $\nabla^2 v$ is diagonal at the point under consideration, Lemma 3.1 implies that

$$(3.17) \quad q_\alpha = -\frac{1}{16} \psi^{-2} \psi_\alpha v_{11} v_{22} v_{33} v_{44} + \frac{1}{16} \psi^{-1} \sum_i \sigma_3(\lambda|i) v_{ii\alpha}.$$

We next estimate ψ_α . Differentiating ψ with respect to x_α and using the diagonal structure of $\nabla^2 v$, we obtain

$$\begin{aligned} \psi_\alpha &= \frac{4}{16^2} (v_{11} + v_{33})(v_{22} + v_{44})^2 (v_{11\alpha} + v_{33\alpha}) + \frac{4}{16^2} (v_{11} + v_{33})^2 (v_{22} + v_{44})(v_{22\alpha} + v_{44\alpha}) \\ &\quad - \frac{2}{16^2} (v_{11} - v_{33})(v_{22} + v_{44})^2 (v_{11\alpha} - v_{33\alpha}) - \frac{2}{16^2} (v_{11} - v_{33})^2 (v_{22} + v_{44})(v_{22\alpha} + v_{44\alpha}) \end{aligned}$$

$$-\frac{2}{16^2}(v_{22}-v_{44})(v_{11}+v_{33})^2(v_{22\alpha}-v_{44\alpha})-\frac{2}{16^2}(v_{22}-v_{44})^2(v_{11}+v_{33})(v_{11\alpha}+v_{33\alpha}).$$

Recall that

$$v_{33} = O(\phi), \quad v_{44} = O(\phi) \text{ and } v_{22} = v_{11} + O(\phi).$$

Substituting these relations into the above expression and collecting the leading-order terms, we obtain

$$\begin{aligned} \psi_\alpha &= \frac{4}{16^2}v_{11}^3(v_{11\alpha}+v_{33\alpha}) + \frac{4}{16^2}v_{11}^3(v_{22\alpha}+v_{44\alpha}) - \frac{2}{16^2}v_{11}^3(v_{11\alpha}-v_{33\alpha}) \\ (3.18) \quad &- \frac{2}{16^2}v_{11}^3(v_{22\alpha}+v_{44\alpha}) - \frac{2}{16^2}v_{11}^3(v_{22\alpha}-v_{44\alpha}) - \frac{2}{16^2}v_{11}^3(v_{11\alpha}+v_{33\alpha}) + O(\phi) \\ &= \frac{4}{16^2}v_{11}^3(v_{33\alpha}+v_{44\alpha}) + O(\phi). \end{aligned}$$

We now estimate the two terms on the right-hand side of (3.17). By (3.16) and (3.18), the first term on the right-hand side of (3.17) satisfies

$$-\frac{1}{16}\psi^{-2}\psi_\alpha v_{11}v_{22}v_{33}v_{44} = -4v_{11}^{-1}\frac{v_{33}v_{44}}{\sigma_1^2(B)}(v_{33\alpha}+v_{44\alpha}) + O(\phi).$$

For the second term, again by (3.16), we have

$$\frac{1}{16}\psi^{-1}\sum_i\sigma_3(\lambda|i)v_{ii\alpha} = 4v_{11}^{-1}\frac{v_{44}}{\sigma_1(B)}v_{33\alpha} + 4v_{11}^{-1}\frac{v_{33}}{\sigma_1(B)}v_{44\alpha} + O(\phi).$$

Combining the above estimates, we conclude that

$$q_\alpha = 4v_{11}^{-1}\frac{v_{44}^2}{\sigma_1^2(B)}v_{33\alpha} + 4v_{11}^{-1}\frac{v_{33}^2}{\sigma_1^2(B)}v_{44\alpha} + O(\phi).$$

This completes the proof. $\square$

Before computing the second-order derivatives of q , we recall the following lemma, which provides an estimate for the third-order derivatives of convex functions. We refer the reader to Bian and Guan [4] for the proof.

Lemma 3.3 (Lemma 2.5 in [4]). *Assume that $v \in C^{3,1}(\Omega)$ is a convex function. Then there exists a positive constant C , depending only on $\text{dist}(\mathcal{O}, \partial\Omega)$ and $\|v\|_{C^{3,1}(\Omega)}$ such that*

$$(3.19) \quad |v_{ij\alpha}(x)| \leq C\left(\sqrt{v_{ii}(x)} + \sqrt{v_{jj}(x)}\right)$$

for all $x \in \mathcal{O}$ and $1 \leq i, j, \alpha \leq n$.

We are now ready to establish the crucial estimates for the second-order derivatives of q .

Proposition 3.4. *Let $q(\nabla^2 v)$ be defined as in (3.6). Assume that $\nabla^2 v$ is diagonal at the point under consideration. Then, for any $\alpha, \beta \in \{1, 2, 3, 4\}$, we have*

$$\begin{aligned} q_{\alpha\beta} = & 4v_{11}^{-1} \frac{v_{44}^2}{\sigma_1^2(B)} v_{33\alpha\beta} - 8v_{11}^{-2} \frac{v_{44}^2}{\sigma_1^2(B)} v_{13\alpha} v_{13\beta} - 8v_{11}^{-2} \frac{v_{44}^2}{\sigma_1^2(B)} v_{23\alpha} v_{23\beta} \\ & + 4v_{11}^{-1} \frac{v_{33}^2}{\sigma_1^2(B)} v_{44\alpha\beta} - 8v_{11}^{-2} \frac{v_{33}^2}{\sigma_1^2(B)} v_{14\alpha} v_{14\beta} - 8v_{11}^{-2} \frac{v_{33}^2}{\sigma_1^2(B)} v_{24\alpha} v_{24\beta} \\ & - 4v_{11}^{-1} \frac{1}{\sigma_1^3(B)} (V_{3\alpha} V_{3\beta} + V_{4\alpha} V_{4\beta}) - 8v_{11}^{-1} \frac{1}{\sigma_1(B)} v_{34\alpha} v_{34\beta} \\ & + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right), \end{aligned}$$

where

$$V_{3\alpha} = \sigma_1(B) v_{33\alpha} - v_{33} \sum_{i \in B} v_{ii\alpha} \quad \text{and} \quad V_{4\alpha} = \sigma_1(B) v_{44\alpha} - v_{44} \sum_{i \in B} v_{ii\alpha}.$$

Proof. Differentiating q twice with respect to x_α and x_β , we have

$$\begin{aligned} 16q_{\alpha\beta} = & 2\psi^{-3} \psi_\alpha \psi_\beta \det(\nabla^2 v) - \psi^{-2} \psi_{\alpha\beta} \det(\nabla^2 v) - \psi^{-2} \psi_\alpha \sum_{i,j} \frac{\partial \det(\nabla^2 v)}{\partial v_{ij}} v_{ij\beta} \\ & - \psi^{-2} \psi_\beta \sum_{i,j} \frac{\partial \det(\nabla^2 v)}{\partial v_{ij}} v_{ij\alpha} + \psi^{-1} \sum_{i,j,k,l} \frac{\partial^2 \det(\nabla^2 v)}{\partial v_{ij} \partial v_{kl}} v_{ij\alpha} v_{kl\beta} \\ & + \psi^{-1} \sum_{i,j} \frac{\partial \det(\nabla^2 v)}{\partial v_{ij}} v_{ij\alpha\beta}. \end{aligned}$$

Since $\nabla^2 v$ is diagonal at the point under consideration, Lemma 3.1 implies that

$$\begin{aligned} 16q_{\alpha\beta} = & 2\psi^{-3} \psi_\alpha \psi_\beta \det(\nabla^2 v) - \psi^{-2} \psi_{\alpha\beta} \det(\nabla^2 v) - \psi^{-2} \psi_\alpha \sum_i \sigma_3(\lambda|i) v_{ii\beta} \\ & - \psi^{-2} \psi_\beta \sum_i \sigma_3(\lambda|i) v_{ii\alpha} + \psi^{-1} \sum_i \sigma_3(\lambda|i) v_{ii\alpha\beta} \\ (3.20) \quad & + \psi^{-1} \sum_{i \neq j} \sigma_2(\lambda|ij) (v_{ii\alpha} v_{jj\beta} - v_{ij\alpha} v_{ji\beta}) \\ & \triangleq \textcircled{1} + \textcircled{2} + \textcircled{3} + \textcircled{4} + \textcircled{5} + \textcircled{6}. \end{aligned}$$

We next estimate the terms $\textcircled{1}$ – $\textcircled{6}$ separately.

• $\textcircled{1}$. By the expansion in (3.16), together with the estimate $\det(\nabla^2 v) = O(\phi^2)$, we have

$$\psi^{-3} \det(\nabla^2 v) = \psi_0^{-3} \det(\nabla^2 v) + O(1).$$

On the other hand, by (3.18), we have

$$\begin{aligned}\psi_\alpha \psi_\beta &= \frac{4^2}{16^4} v_{11}^6 (v_{33\alpha} + v_{44\alpha})(v_{33\beta} + v_{44\beta}) \\ &\quad + \frac{4}{16^2} v_{11}^3 (v_{33\alpha} + v_{44\alpha} + v_{33\beta} + v_{44\beta}) \cdot O(\phi) + O(\phi^2).\end{aligned}$$

Combining these estimates, we obtain

$$\begin{aligned}2\psi^{-3}\psi_\alpha\psi_\beta\det(\nabla^2v) &= (2\psi_0^{-3}\det(\nabla^2v) + O(1)) \cdot \left(\frac{4^2}{16^4} v_{11}^6 (v_{33\alpha} + v_{44\alpha})(v_{33\beta} + v_{44\beta}) \right. \\ &\quad \left. + \frac{4}{16^2} v_{11}^3 (v_{33\alpha} + v_{44\alpha} + v_{33\beta} + v_{44\beta}) \cdot O(\phi) + O(\phi^2) \right).\end{aligned}$$

By Lemma 3.3, for any $p, q \in B$ and $\alpha \in \{1, 2, 3, 4\}$, we have

$$(3.21) \quad v_{pq\alpha} = O(\sqrt{\phi}).$$

Therefore, the above estimates yield

$$\begin{aligned}2\psi^{-3}\psi_\alpha\psi_\beta\det(\nabla^2v) &= 2\psi_0^{-3}\det(\nabla^2v) \cdot \frac{4^2}{16^4} v_{11}^6 (v_{33\alpha} + v_{44\alpha})(v_{33\beta} + v_{44\beta}) + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right) \\ &= 8 \cdot 16v_{11}^{-1} \frac{\sigma_2(B)}{\sigma_1^3(B)} (v_{33\alpha} + v_{44\alpha})(v_{33\beta} + v_{44\beta}) + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right).\end{aligned}$$

- ②. Following the same argument used to estimate ①, we obtain

$$\begin{aligned}-\psi^{-2}\det(\nabla^2v) &= -\psi_0^{-2}\det(\nabla^2v) + O(\phi) \\ &= -16^3 v_{11}^{-4} \frac{\sigma_2(B)}{\sigma_1^2(B)} + O(\phi).\end{aligned}$$

We now turn to the computation of $\psi_{\alpha\beta}$. Recall the explicit expression for ψ in (3.7). Since ∇^2v is diagonal at the point under consideration, differentiating this expression twice yields

$$\begin{aligned}(3.22) \quad \psi_{\alpha\beta} &= \frac{4}{16^2} v_{11}^3 v_{33\alpha\beta} - \frac{8}{16^2} v_{11}^2 v_{13\alpha} v_{13\beta} - \frac{8}{16^2} v_{11}^2 v_{23\alpha} v_{23\beta} \\ &\quad + \frac{4}{16^2} v_{11}^3 v_{44\alpha\beta} - \frac{8}{16^2} v_{11}^2 v_{14\alpha} v_{14\beta} - \frac{8}{16^2} v_{11}^2 v_{24\alpha} v_{24\beta} \\ &\quad + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right).\end{aligned}$$

Here we have repeatedly used the estimates (3.12), (3.13) and (3.21). Thus,

$$\begin{aligned}-\psi^{-2}\psi_{\alpha\beta}\det(\nabla^2v) &= -4 \cdot 16v_{11}^{-1} \frac{\sigma_2(B)}{\sigma_1^2(B)} v_{33\alpha\beta} + 8 \cdot 16v_{11}^{-2} \frac{\sigma_2(B)}{\sigma_1^2(B)} v_{13\alpha} v_{13\beta} \\ &\quad + 8 \cdot 16v_{11}^{-2} \frac{\sigma_2(B)}{\sigma_1^2(B)} v_{23\alpha} v_{23\beta} - 4 \cdot 16v_{11}^{-1} \frac{\sigma_2(B)}{\sigma_1^2(B)} v_{44\alpha\beta}\end{aligned}$$

$$\begin{aligned}
& + 8 \cdot 16v_{11}^{-2} \frac{\sigma_2(B)}{\sigma_1^2(B)} v_{14\alpha} v_{14\beta} + 8 \cdot 16v_{11}^{-2} \frac{\sigma_2(B)}{\sigma_1^2(B)} v_{24\alpha} v_{24\beta} \\
& + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right).
\end{aligned}$$

• ③ and ④. For the term ③, combining (3.16) and (3.18) with the estimates (3.12), (3.13) and (3.21), we obtain

$$\begin{aligned}
-\psi^{-2} \psi_\alpha \sigma_3(\lambda|i) v_{ii\beta} &= -(\psi_0^{-2} + O(\phi^{-1})) \cdot \left(\frac{4}{16^2} v_{11}^3 (v_{33\alpha} + v_{44\alpha}) + O(\phi) \right) \\
&\quad \cdot \left(v_{11}^2 v_{44} v_{33\beta} + v_{11}^2 v_{33} v_{44\beta} + O(\phi^2) \right) \\
&= -\frac{1}{\frac{4^2}{16^4} v_{11}^6 (v_{33} + v_{44})^2} \cdot \frac{4}{16^2} v_{11}^3 (v_{33\alpha} + v_{44\alpha}) \cdot (v_{11}^2 v_{44} v_{33\beta} + v_{11}^2 v_{33} v_{44\beta}) \\
&\quad + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right) \\
&= -4 \cdot 16v_{11}^{-1} \frac{v_{44}}{\sigma_1^2(B)} (v_{33\alpha} + v_{44\alpha}) v_{33\beta} - 4 \cdot 16v_{11}^{-1} \frac{v_{33}}{\sigma_1^2(B)} (v_{33\alpha} + v_{44\alpha}) v_{44\beta} \\
&\quad + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right).
\end{aligned}$$

The term ④ admits an analogous estimate. More precisely, exchanging the roles of α and β in the above computation gives

$$\begin{aligned}
-\psi^{-2} \psi_\beta \sigma_3(\lambda|i) v_{ii\alpha} &= -4 \cdot 16v_{11}^{-1} \frac{v_{44}}{\sigma_1^2(B)} (v_{33\beta} + v_{44\beta}) v_{33\alpha} - 4 \cdot 16v_{11}^{-1} \frac{v_{33}}{\sigma_1^2(B)} (v_{33\beta} + v_{44\beta}) v_{44\alpha} \\
&\quad + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}|\right).
\end{aligned}$$

• ⑤. A direct computation gives

$$\begin{aligned}
\psi^{-1} \sigma_3(\lambda|i) v_{ii\alpha\beta} &= (\psi_0^{-1} + O(1)) \cdot (v_{11}^2 v_{44} v_{33\alpha\beta} + v_{11}^2 v_{33} v_{44\alpha\beta} + O(\phi^2)) \\
&= 4 \cdot 16v_{11}^{-1} \frac{v_{44}}{\sigma_1(B)} v_{33\alpha\beta} + 4 \cdot 16v_{11}^{-1} \frac{v_{33}}{\sigma_1(B)} v_{44\alpha\beta} + O(\phi).
\end{aligned}$$

• ⑥. Using the expansion in (3.16), together with the estimates (3.12), (3.13) and (3.21), we have

$$\begin{aligned}
&\psi^{-1} \sum_{i \neq j} \sigma_2(\lambda|ij) (v_{ii\alpha} v_{jj\beta} - v_{ij\alpha} v_{ji\beta}) \\
&= (\psi_0^{-1} + O(1)) \cdot [v_{33} v_{44} (v_{11\alpha} v_{22\beta} - v_{12\alpha} v_{21\beta}) + v_{33} v_{44} (v_{22\alpha} v_{11\beta} - v_{21\alpha} v_{12\beta})]
\end{aligned}$$

$$\begin{aligned}
& + v_{22}v_{44}(v_{11\alpha}v_{33\beta} - v_{13\alpha}v_{31\beta}) + v_{22}v_{44}(v_{33\alpha}v_{11\beta} - v_{31\alpha}v_{13\beta}) \\
& + v_{22}v_{33}(v_{11\alpha}v_{44\beta} - v_{14\alpha}v_{41\beta}) + v_{22}v_{33}(v_{44\alpha}v_{11\beta} - v_{41\alpha}v_{14\beta}) \\
& + v_{11}v_{44}(v_{22\alpha}v_{33\beta} - v_{23\alpha}v_{32\beta}) + v_{11}v_{44}(v_{33\alpha}v_{22\beta} - v_{32\alpha}v_{23\beta}) \\
& + v_{11}v_{33}(v_{22\alpha}v_{44\beta} - v_{24\alpha}v_{42\beta}) + v_{11}v_{33}(v_{44\alpha}v_{22\beta} - v_{42\alpha}v_{24\beta}) \\
& + v_{11}v_{22}(v_{33\alpha}v_{44\beta} - v_{34\alpha}v_{43\beta}) + v_{11}v_{22}(v_{44\alpha}v_{33\beta} - v_{43\alpha}v_{34\beta}) \Big] \\
= & \psi_0^{-1} \cdot \left(-2v_{22}v_{44}v_{13\alpha}v_{13\beta} - 2v_{22}v_{33}v_{14\alpha}v_{14\beta} - 2v_{11}v_{44}v_{23\alpha}v_{23\beta} \right. \\
& - 2v_{11}v_{33}v_{24\alpha}v_{24\beta} + v_{11}v_{22}v_{33\alpha}v_{44\beta} + v_{11}v_{22}v_{44\alpha}v_{33\beta} \\
& \left. - 2v_{11}v_{22}v_{34\alpha}v_{34\beta} \right) + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}| \right) \\
= & -8 \cdot 16v_{11}^{-2} \frac{v_{44}}{\sigma_1(B)} v_{13\alpha}v_{13\beta} - 8 \cdot 16v_{11}^{-2} \frac{v_{33}}{\sigma_1(B)} v_{14\alpha}v_{14\beta} - 8 \cdot 16v_{11}^{-2} \frac{v_{44}}{\sigma_1(B)} v_{23\alpha}v_{23\beta} \\
& - 8 \cdot 16v_{11}^{-2} \frac{v_{33}}{\sigma_1(B)} v_{24\alpha}v_{24\beta} + 4 \cdot 16v_{11}^{-1} \frac{1}{\sigma_1(B)} v_{33\alpha}v_{44\beta} + 4 \cdot 16v_{11}^{-1} \frac{1}{\sigma_1(B)} v_{44\alpha}v_{33\beta} \\
& - 8 \cdot 16v_{11}^{-1} \frac{1}{\sigma_1(B)} v_{34\alpha}v_{34\beta} + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}| \right).
\end{aligned}$$

Substituting the estimates obtained above for ①–⑥ into (3.20), we obtain

$$\begin{aligned}
16q_{\alpha\beta} = & 4 \cdot 16v_{11}^{-1} \frac{v_{44}^2}{\sigma_1^2(B)} v_{33\alpha\beta} - 8 \cdot 16v_{11}^{-2} \frac{v_{44}^2}{\sigma_1^2(B)} v_{13\alpha}v_{13\beta} - 8 \cdot 16v_{11}^{-2} \frac{v_{44}^2}{\sigma_1^2(B)} v_{23\alpha}v_{23\beta} \\
& + 4 \cdot 16v_{11}^{-1} \frac{v_{33}^2}{\sigma_1^2(B)} v_{44\alpha\beta} - 8 \cdot 16v_{11}^{-2} \frac{v_{33}^2}{\sigma_1^2(B)} v_{14\alpha}v_{14\beta} - 8 \cdot 16v_{11}^{-2} \frac{v_{33}^2}{\sigma_1^2(B)} v_{24\alpha}v_{24\beta} \\
& - 4 \cdot 16v_{11}^{-1} \frac{1}{\sigma_1^3(B)} (V_{3\alpha}V_{3\beta} + V_{4\alpha}V_{4\beta}) - 8 \cdot 16v_{11}^{-1} \frac{1}{\sigma_1(B)} v_{34\alpha}v_{34\beta} \\
& + O\left(\phi + \sum_{p,q \in B} |\nabla v_{pq}| \right),
\end{aligned}$$

where

$$V_{3\alpha} = \sigma_1(B)v_{33\alpha} - v_{33} \sum_{i \in B} v_{ii\alpha}, \text{ and } V_{4\alpha} = \sigma_1(B)v_{44\alpha} - v_{44} \sum_{i \in B} v_{ii\alpha}.$$

Dividing both sides by the common factor 16, we complete the proof of the proposition. $\square$

Remark 3.5. *In the present setting, we have the simplification*

$$V_{3\alpha} = v_{44}v_{33\alpha} - v_{33}v_{44\alpha} = -V_{4\alpha}, \quad V_{3\beta} = v_{44}v_{33\beta} - v_{33}v_{44\beta} = -V_{4\beta}.$$

Consequently,

$$V_{3\alpha}V_{3\beta} + V_{4\alpha}V_{4\beta} = 2V_{3\alpha}V_{3\beta} = 2(v_{44}v_{33\alpha} - v_{33}v_{44\alpha})(v_{44}v_{33\beta} - v_{33}v_{44\beta}).$$

To establish the differential inequality in the next subsection, we need the following lemma due to Bian and Guan [4]. The proof can be found in their original work.

Lemma 3.6 (Lemma 3.3 in [4]). *Let M be a positive constant such that $0 < \lambda_i \leq M$ and $\frac{1}{M} \leq \gamma_i \leq M$ for $i = l+1, l+2, \dots, n$. Assume that $v_{ij\alpha} = v_{ji\alpha}$ for $i, j = l+1, l+2, \dots, n$ and $\alpha = 1, 2, \dots, n$. Then there exists a positive constant C depending only on n and M , such that for each α , for any $D > 0$ and any $\delta > 0$, we have*

$$\begin{aligned} \sum_{l+1 \leq i, j \leq n} |v_{ij\alpha}| &\leq C \left(1 + \frac{2D}{\delta} + D \right) \left(\sigma_1(\lambda) + \left| \sum_{l+1 \leq i \leq n} \gamma_i v_{ii\alpha} \right| \right) \\ &\quad + \frac{\delta}{2D} \frac{1}{\sigma_1(\lambda)} \sum_{\substack{l+1 \leq i, j \leq n \\ i \neq j}} |v_{ij\alpha}|^2 + \frac{C}{D} \frac{1}{\sigma_1^3(\lambda)} \sum_{l+1 \leq i \leq n} V_{i\alpha}^2, \end{aligned}$$

where

$$\sigma_1(\lambda) = \sum_{l+1 \leq i \leq n} \lambda_i \quad \text{and} \quad V_{i\alpha} = \sigma_1(\lambda) v_{ii\alpha} - \lambda_i \sum_{l+1 \leq j \leq n} v_{jj\alpha}.$$

3.2. A differential inequality. The purpose of this subsection is to derive a differential inequality satisfied by the auxiliary function ϕ defined in (3.3) for equation (2.6). This inequality is a key ingredient in the proof of the constant rank theorem.

Theorem 3.7. *Suppose that $\Omega \subset \mathbb{R}^4$ is a domain and $v \in C^4(\Omega)$ is a convex solution of equation (2.6). Let ϕ be the function defined in (3.3). Namely, for each $x \in \Omega$, set*

$$\phi(x) = \text{tr}(K(x)) + q(K(x)),$$

where

$$q(K(x)) = \begin{cases} \frac{\det(K(x))}{\text{tr}(K(x))} & \text{if } \text{tr}(K(x)) > 0 \\ 0 & \text{if } \text{tr}(K(x)) = 0. \end{cases}$$

If the Hessian $\nabla^2 v(x)$ attains the minimal rank 2 at some point $x_0 \in \Omega$, then there exist a neighborhood $\mathcal{O}$ of x_0 and a positive constant C independent of ϕ such that

$$(3.23) \quad \sum_{i,j} F^{ij} \phi_{ij} \leq C(\phi + |\nabla \phi|) \quad \text{in } \mathcal{O}.$$

Proof. As in the previous subsection, we introduce the perturbation $v_\varepsilon(x) = v(x) + \frac{\varepsilon}{2}|x|^2$ to ensure that $\text{tr}(K_\varepsilon(x))$ remains strictly positive. We denote by K_ε the matrix K associated with v_ε . Then the function v_ε satisfies equation

$$(3.24) \quad F(\nabla^2 v_\varepsilon(x), \nabla v_\varepsilon(x), v_\varepsilon(x)) = R_\varepsilon(x),$$

where $R_\varepsilon = F(\nabla^2 v_\varepsilon, \nabla v_\varepsilon, v_\varepsilon) - F(\nabla^2 v, \nabla v, v)$ and $F(\nabla^2 v, \nabla v, v)$ is defined as in (2.6). Since $v \in C^4(\Omega)$, for any $x \in \mathcal{O}$, we have

$$(3.25) \quad |R_\varepsilon(x)| \leq C\varepsilon, \quad |\nabla R_\varepsilon(x)| \leq C\varepsilon \quad \text{and} \quad |\nabla^2 R_\varepsilon(x)| \leq C\varepsilon.$$

Recalling (3.11), we know that

$$(3.26) \quad \text{tr}(K_\varepsilon(x)) \geq c\varepsilon \quad \text{for all } x \in \mathcal{O}.$$

Here the positive constants C and c are independent of ε . In view of (3.5), we define

$$\phi_\varepsilon(x) = (\det(A_\varepsilon))^{-2}\psi_\varepsilon(x) + q_\varepsilon(\nabla^2 v_\varepsilon(x)),$$

where

$$q_\varepsilon(\nabla^2 v_\varepsilon(x)) = \frac{\det(\nabla^2 v_\varepsilon(x))}{16\psi_\varepsilon(x)}.$$

Here A_ε and ψ_ε are defined by (2.13) and (3.7), respectively, with v replaced by v_ε . Applying the argument developed for equation (3.24), we will establish a differential inequality for ϕ_ε analogous to (3.23). By letting $\varepsilon \rightarrow 0$, we obtain (3.23).

In the following, we continue to denote v_ε , ∇v_ε , $\nabla^2 v_\varepsilon$, R_ε , A_ε , K_ε , ψ_ε , ϕ_ε , and so on, simply by v , ∇v , $\nabla^2 v$, R , A , K , ψ , ϕ , respectively, with the understanding that all estimates are independent of ε . With this convention, the perturbed function v satisfies

$$(3.27) \quad F(\nabla^2 v, \nabla v, v) = R(x),$$

where F is defined as in (2.6).

Fix a point $x \in \mathcal{O}$ and choose the coordinates described in Section 2.2. We may assume that $\nabla^2 v(x)$ is diagonal and satisfies

$$v_{11} + v_{33} = v_{22} + v_{44},$$

with the eigenvalues ordered as $v_{11}(x) \geq v_{33}(x) \geq \varepsilon$ and $v_{22}(x) \geq v_{44}(x) \geq \varepsilon$. By (3.26), we have $\varepsilon \leq C\phi(x)$. Combining this estimate with (3.25), we obtain

$$(3.28) \quad R(x) = O(\phi(x)), \quad \nabla R(x) = O(\phi(x)) \quad \text{and} \quad \nabla^2 R(x) = O(\phi(x)).$$

In what follows, all computations are carried out at the fixed point x . Since the minimal rank of $\nabla^2 v$ is 2, we know that $v_{11} \geq c$ and $v_{22} \geq c$ for some positive constant c . Let us divide the indices into the “good” and “bad” sets, denoted by $G = \{1, 2\}$ and $B = \{3, 4\}$, respectively. For later convenience, we restate the previously obtained estimates (3.12) and (3.13) in the form

$$(3.29) \quad v_{33} = O(\phi), \quad v_{44} = O(\phi)$$

and

$$(3.30) \quad v_{22} = v_{11} + O(\phi).$$

From the analysis in Section 3.1, we know that $\phi = (\det(A))^{-2}\psi + q$, where q is the quotient component of ϕ and

$$\det(A) = \frac{1}{16}((v_{11} + v_{33})(v_{22} + v_{44}) - (v_{12} + v_{34})(v_{21} + v_{43}) - (v_{14} - v_{32})(v_{41} - v_{23})).$$

Taking the first derivatives of ϕ at the fixed point x , and applying Lemma 3.2, we obtain

$$\begin{aligned} \phi_i = & -2 \cdot 16^2 (v_{11} + v_{33})^{-3} (v_{22} + v_{44})^{-3} \\ & \cdot [(v_{11i} + v_{33i})(v_{22} + v_{44}) + (v_{11} + v_{33})(v_{22i} + v_{44i})] \psi \\ & + 16^2 (v_{11} + v_{33})^{-2} (v_{22} + v_{44})^{-2} \psi_i \end{aligned}$$

$$+ 4v_{11}^{-1} \frac{v_{44}^2}{\sigma_1^2(B)} v_{33i} + 4v_{11}^{-1} \frac{v_{33}^2}{\sigma_1^2(B)} v_{44i} + O(\phi)$$

Note that $v_{11} + v_{33}$ and $v_{22} + v_{44}$ are bounded away from zero. By the estimates $\psi = O(\phi)$ and (3.18), we have

$$(3.31) \quad \phi_i = 4v_{11}^{-1} \left(1 + \frac{v_{44}^2}{\sigma_1^2(B)} \right) v_{33i} + 4v_{11}^{-1} \left(1 + \frac{v_{33}^2}{\sigma_1^2(B)} \right) v_{44i} + O(\phi).$$

Recalling (3.15) and (3.18), we have

$$\psi = O(\phi) \quad \text{and} \quad \psi_i = O\left(\phi + \sum_{\alpha, \beta \in B} |\nabla v_{\alpha\beta}|\right).$$

It follows from (3.22) and Proposition 3.4 that the second-order derivatives of ϕ at x are given by

$$\begin{aligned} \phi_{ij} = & 4v_{11}^{-1} \left(1 + \frac{v_{44}^2}{\sigma_1^2(B)} \right) v_{33ij} - 8v_{11}^{-2} \left(1 + \frac{v_{44}^2}{\sigma_1^2(B)} \right) v_{13i} v_{13j} \\ & - 8v_{11}^{-2} \left(1 + \frac{v_{44}^2}{\sigma_1^2(B)} \right) v_{23i} v_{23j} + 4v_{11}^{-1} \left(1 + \frac{v_{33}^2}{\sigma_1^2(B)} \right) v_{44ij} \\ & - 8v_{11}^{-2} \left(1 + \frac{v_{33}^2}{\sigma_1^2(B)} \right) v_{14i} v_{14j} - 8v_{11}^{-2} \left(1 + \frac{v_{33}^2}{\sigma_1^2(B)} \right) v_{24i} v_{24j} \\ & - 4v_{11}^{-1} \frac{1}{\sigma_1^3(B)} (V_{3i} V_{3j} + V_{4i} V_{4j}) - 8v_{11}^{-1} \frac{1}{\sigma_1(B)} v_{34i} v_{34j} \\ & + O\left(\phi + \sum_{\alpha, \beta \in B} |\nabla v_{\alpha\beta}|\right), \end{aligned}$$

where

$$V_{3i} = \sigma_1(B) v_{33i} - v_{33} \sum_{\gamma \in B} v_{\gamma\gamma i}, \quad \text{and} \quad V_{4i} = \sigma_1(B) v_{44i} - v_{44} \sum_{\gamma \in B} v_{\gamma\gamma i}.$$

Thus,

$$\begin{aligned}
(3.32) \quad \sum_{i,j} F^{ij} \phi_{ij} &= 4v_{11}^{-1} \sum_{i,j} \sum_{\alpha \in B} \left(1 + \frac{\sigma_1^2(B|\alpha)}{\sigma_1^2(B)}\right) F^{ij} v_{\alpha\alpha ij} \\
&\quad - 8v_{11}^{-2} \sum_{i,j} \sum_{\alpha \in B, \beta \in G} \left(1 + \frac{\sigma_1^2(B|\alpha)}{\sigma_1^2(B)}\right) F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \\
&\quad - 4v_{11}^{-1} \frac{1}{\sigma_1^3(B)} \sum_{i,j} \sum_{\alpha \in B} F^{ij} V_{\alpha i} V_{\alpha j} \\
&\quad - 4v_{11}^{-1} \frac{1}{\sigma_1(B)} \sum_{i,j} \sum_{\substack{\alpha, \beta \in B \\ \alpha \neq \beta}} F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \\
&\quad + O\left(\phi + \sum_{\alpha, \beta \in B} |\nabla v_{\alpha\beta}|\right),
\end{aligned}$$

where

$$V_{\alpha i} = \sigma_1(B) v_{\alpha\alpha i} - v_{\alpha\alpha} \sum_{\gamma \in B} v_{\gamma\gamma i}.$$

To handle the terms involving fourth-order derivatives in (3.32), we differentiate equation (3.27) once to obtain

$$(3.33) \quad \sum_{i,j} F^{ij} v_{ij\alpha} + \sum_k F^{v_k} v_{k\alpha} + F^v v_\alpha = O(\phi),$$

and differentiate it a second time to obtain

$$\begin{aligned}
(3.34) \quad O(\phi) &= \sum_{i,j,k,l} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + \sum_{i,j,k} F^{ij,v_k} v_{ij\alpha} v_{k\alpha} + \sum_{i,j} F^{ij,v} v_{ij\alpha} v_\alpha + \sum_{i,j} F^{ij} v_{ij\alpha\alpha} \\
&\quad + \sum_{k,i,j} F^{v_k,ij} v_{k\alpha} v_{ij\alpha} + \sum_{k,l} F^{v_k,v_l} v_{k\alpha} v_{l\alpha} + \sum_k F^{v_k,v} v_{k\alpha} v_\alpha + \sum_k F^{v_k} v_{k\alpha\alpha} \\
&\quad + \sum_{i,j} F^{v,ij} v_\alpha v_{ij\alpha} + \sum_k F^{v,v_k} v_\alpha v_{k\alpha} + F^{v,v} v_\alpha v_\alpha + F^v v_{\alpha\alpha},
\end{aligned}$$

where we have used the estimates in (3.28). Therefore, combining (3.34) with the estimates (3.29), (3.30) and Lemma 3.3, we obtain

$$\begin{aligned}
& 4v_{11}^{-1} \sum_{i,j} \sum_{\alpha \in B} \left(1 + \frac{\sigma_1^2(B|\alpha)}{\sigma_1^2(B)} \right) F^{ij} v_{\alpha\alpha ij} \\
&= -4v_{11}^{-1} \sum_{\alpha \in B} \left(1 + \frac{\sigma_1^2(B|\alpha)}{\sigma_1^2(B)} \right) \\
&\quad \cdot \left(\sum_{i,j,k,l} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j,k} F^{ij,v_k} v_{ij\alpha} v_{k\alpha} + 2 \sum_{i,j} F^{ij,v} v_{ij\alpha} v_{\alpha} \right. \\
&\quad \left. + \sum_{k,l} F^{v_k,v_l} v_{k\alpha} v_{l\alpha} + 2 \sum_k F^{v_k,v} v_{k\alpha} v_{\alpha} + F^{v,v} v_{\alpha}^2 + \sum_k F^{v_k} v_{k\alpha\alpha} \right. \\
&\quad \left. + F^v v_{\alpha\alpha} + O(\phi) \right) \\
&= -4v_{11}^{-1} \sum_{\alpha \in B} \left(1 + \frac{\sigma_1^2(B|\alpha)}{\sigma_1^2(B)} \right) \\
&\quad \cdot \left(\sum_{i,j,k,l \in G} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij\alpha} v_{\alpha} + F^{v,v} v_{\alpha}^2 \right) \\
&\quad + O\left(\phi + \sum_{\alpha, \beta \in B} |\nabla v_{\alpha\beta}| \right).
\end{aligned} \tag{3.35}$$

Substituting (3.35) into (3.32) yields

$$\begin{aligned}
\sum_{i,j} F^{ij} \phi_{ij} &= -4v_{11}^{-1} \sum_{\alpha \in B} \left(1 + \frac{\sigma_1^2(B|\alpha)}{\sigma_1^2(B)} \right) \\
&\quad \cdot \left(\sum_{i,j,k,l \in G} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij\alpha} v_{\alpha} + F^{v,v} v_{\alpha}^2 \right. \\
&\quad \left. + 2v_{11}^{-1} \sum_{i,j,\beta \in G} F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \right) \\
&\quad - 4v_{11}^{-1} \left(\frac{1}{\sigma_1^3(B)} \sum_{i,j} \sum_{\alpha \in B} F^{ij} V_{\alpha i} V_{\alpha j} + \frac{1}{\sigma_1(B)} \sum_{i,j} \sum_{\substack{\alpha, \beta \in B \\ \alpha \neq \beta}} F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \right) \\
&\quad + O\left(\phi + \sum_{\alpha, \beta \in B} |\nabla v_{\alpha\beta}| \right).
\end{aligned} \tag{3.36}$$

To establish the differential inequality (3.23), we first prove the following claim.

Claim 1. For each $\alpha \in B$, we have

$$(3.37) \quad \begin{aligned} & \sum_{i,j,k,l \in G} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij\alpha} v_{\alpha} + F^{v,v} v_{\alpha}^2 + 2 \sum_{i,j,\beta \in G} \frac{1}{v_{\beta\beta}} F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \\ & \geq -C \left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}| \right). \end{aligned}$$

Once **Claim 1** is proved, since $v_{22} = v_{11} + O(\phi)$, then (3.36) gives

$$\begin{aligned} \sum_{i,j} F^{ij} \phi_{ij} & \leq -4v_{11}^{-1} \left(\frac{1}{\sigma_1^3(B)} \sum_{i,j} \sum_{\alpha \in B} F^{ij} V_{\alpha i} V_{\alpha j} + \frac{1}{\sigma_1(B)} \sum_{i,j} \sum_{\substack{\alpha,\beta \in B \\ \alpha \neq \beta}} F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \right) \\ & \quad + C \left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}| \right), \end{aligned}$$

where

$$V_{\alpha i} = \sigma_1(B) v_{\alpha\alpha i} - v_{\alpha\alpha} \sum_{\gamma \in B} v_{\gamma\gamma i}.$$

Since $v \in C^4(\Omega)$ and $\bar{\mathcal{O}} \subset \Omega$, there exists a constant $\delta_0 > 0$ such that

$$(F^{ij}) \geq \delta_0 I_4 \text{ in } \mathcal{O}.$$

Consequently, we can deduce

$$\sum_{i,j} F^{ij} V_{\alpha i} V_{\alpha j} \geq \delta_0 \sum_i V_{\alpha i}^2 \text{ and } \sum_{i,j} F^{ij} v_{\alpha\beta i} v_{\alpha\beta j} \geq \delta_0 \sum_i v_{\alpha\beta i}^2.$$

It follows that

$$\sum_{i,j} F^{ij} \phi_{ij} \leq -\frac{\delta'_0}{\sigma_1^3(B)} \sum_i \sum_{\alpha \in B} V_{\alpha i}^2 - \frac{\delta'_0}{\sigma_1(B)} \sum_i \sum_{\substack{\alpha,\beta \in B \\ \alpha \neq \beta}} v_{\alpha\beta i}^2 + C \left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}| \right).$$

By Lemma 3.6 and the estimate (3.31), we finally obtain

$$\sum_{i,j} F^{ij} \phi_{ij} \leq C (\phi + |\nabla \phi|).$$

Since the constant C is independent of ε , letting $\varepsilon \rightarrow 0$ yields (3.23) for v . $\square$

Proof of Claim 1. Recall the definition of F , namely,

$$\begin{aligned} F(\nabla^2 v, \nabla v, v) & = v^2 [(v_{11} + v_{33})(v_{22} + v_{44}) - (v_{12} + v_{34})(v_{21} + v_{43}) - (v_{14} - v_{32})(v_{41} - v_{23})] \\ & \quad + v [(v_2^2 + v_4^2)(v_{11} + v_{33}) + (v_1^2 + v_3^2)(v_{22} + v_{44}) \\ & \quad - (v_1 v_2 + v_3 v_4)(v_{12} + v_{34} + v_{21} + v_{43}) \\ & \quad - (v_1 v_4 - v_2 v_3)(v_{14} - v_{32} + v_{41} - v_{23})] - 1. \end{aligned}$$

It is clear that

$$F^{11,22} = F^{22,11} = v^2, \quad F^{12,21} = F^{21,12} = -v^2,$$

$$F^{11,v} = 2v(v_{22} + v_{44}) + v_2^2 + v_4^2 = \frac{F^{11}}{v} + v(v_{22} + v_{44}),$$

$$F^{22,v} = 2v(v_{11} + v_{33}) + v_1^2 + v_3^2 = \frac{F^{22}}{v} + v(v_{11} + v_{33}),$$

and

$$F^{12,v} = -(v_1v_2 + v_3v_4) = \frac{F^{12}}{v}, \quad F^{v,v} = 2(v_{11} + v_{33})(v_{22} + v_{44}).$$

Note that $G = \{1, 2\}$ and $B = \{3, 4\}$. Expanding and collecting the terms on the left-hand side of (3.37), we obtain

$$(3.38) \quad \begin{aligned} & \sum_{i,j,k,l \in G} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij\alpha} v_\alpha + F^{v,v} v_\alpha^2 + 2 \sum_{i,j,\beta \in G} \frac{1}{v_{\beta\beta}} F^{ij} v_{\alpha\beta i} v_{\beta\alpha j} \\ &= \frac{2F^{11}}{v_{11}} v_{11\alpha}^2 + \frac{2}{v_{11}v_{22}} (F^{11}v_{11} + F^{22}v_{22} - v^2v_{11}v_{22}) v_{12\alpha}^2 + \frac{2F^{22}}{v_{22}} v_{22\alpha}^2 + 4 \frac{F^{12}}{v_{11}} v_{11\alpha} v_{12\alpha} \\ & \quad + 2v^2 v_{11\alpha} v_{22\alpha} + 4 \frac{F^{12}}{v_{22}} v_{12\alpha} v_{22\alpha} + 2 \left(\frac{F^{11}}{v} + v(v_{22} + v_{44}) \right) v_{11\alpha} v_\alpha \\ & \quad + 4 \frac{F^{12}}{v} v_{12\alpha} v_\alpha + 2 \left(\frac{F^{22}}{v} + v(v_{11} + v_{33}) \right) v_{22\alpha} v_\alpha + 2(v_{11} + v_{33})(v_{22} + v_{44}) v_\alpha^2. \end{aligned}$$

By (3.29), equation (3.33) is equivalent to

$$\sum_{i,j \in G} F^{ij} v_{ij\alpha} + F^v v_\alpha = O\left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}|\right),$$

namely,

$$(3.39) \quad v_\alpha = -\frac{F^{11}}{F^v} v_{11\alpha} - 2 \frac{F^{12}}{F^v} v_{12\alpha} - \frac{F^{22}}{F^v} v_{22\alpha} + O\left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}|\right).$$

In equation (3.39), we used the fact that $F^v \neq 0$. This can be derived from the identity

$$(3.40) \quad F^{11}v_{11} + F^{22}v_{22} + F^{33}v_{33} + F^{44}v_{44} = vF^v.$$

Substituting (3.39) into (3.38), we have

$$\begin{aligned}
& \sum_{i,j,k,l \in G} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij\alpha} v_{\alpha} + F^{v,v} v_{\alpha}^2 + 2 \sum_{i,j,\beta \in G} \frac{1}{v_{\beta\beta}} F^{ij} v_{\alpha\beta i} v_{\beta\alpha j} \\
&= \left(\frac{2F^{11}}{v_{11}} - 2 \left(\frac{F^{11}}{v} + vv_{22} \right) \frac{F^{11}}{F^v} + 2v_{11}v_{22} \left(\frac{F^{11}}{F^v} \right)^2 \right) v_{11\alpha}^2 \\
&+ \left(\frac{2}{v_{11}v_{22}} (F^{11}v_{11} + F^{22}v_{22} - v^2v_{11}v_{22}) - \frac{8(F^{12})^2}{vF^v} + 8v_{11}v_{22} \left(\frac{F^{12}}{F^v} \right)^2 \right) v_{12\alpha}^2 \\
&+ \left(\frac{2F^{22}}{v_{22}} - 2 \left(\frac{F^{22}}{v} + vv_{11} \right) \frac{F^{22}}{F^v} + 2v_{11}v_{22} \left(\frac{F^{22}}{F^v} \right)^2 \right) v_{22\alpha}^2 \\
&+ 2 \left(\frac{2F^{12}}{v_{11}} - 2 \left(\frac{F^{11}}{v} + vv_{22} \right) \frac{F^{12}}{F^v} - \frac{2F^{11}F^{12}}{vF^v} + 4v_{11}v_{22} \frac{F^{11}F^{12}}{(F^v)^2} \right) v_{11\alpha}v_{12\alpha} \\
&+ 2 \left(v^2 - \left(\frac{F^{11}}{v} + vv_{22} \right) \frac{F^{22}}{F^v} - \left(\frac{F^{22}}{v} + vv_{11} \right) \frac{F^{11}}{F^v} + 2v_{11}v_{22} \frac{F^{11}F^{22}}{(F^v)^2} \right) v_{11\alpha}v_{22\alpha} \\
&+ 2 \left(\frac{2F^{12}}{v_{22}} - 2 \left(\frac{F^{22}}{v} + vv_{11} \right) \frac{F^{12}}{F^v} - \frac{2F^{12}F^{22}}{vF^v} + 4v_{11}v_{22} \frac{F^{12}F^{22}}{(F^v)^2} \right) v_{12\alpha}v_{22\alpha} \\
&+ O \left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}| \right).
\end{aligned} \tag{3.41}$$

Recalling our equation $F(\nabla^2 v, \nabla v, v) = R(x)$ in (3.27), and utilizing the estimates $v_{33} = O(\phi)$ and $v_{44} = O(\phi)$, we obtain

$$v^2 v_{11} v_{22} + v(v_2^2 + v_4^2) v_{11} + v(v_1^2 + v_3^2) v_{22} = 1 + O(\phi),$$

and

$$F^{11} = v^2 v_{22} + v(v_2^2 + v_4^2) + O(\phi), \quad F^{22} = v^2 v_{11} + v(v_1^2 + v_3^2) + O(\phi),$$

$$F^v = 2v v_{11} v_{22} + (v_2^2 + v_4^2) v_{11} + (v_1^2 + v_3^2) v_{22} + O(\phi).$$

Clearly, F^{11} , F^{22} and F^v satisfy the following relations

$$(3.42) \quad vF^v = F^{11}v_{11} + F^{22}v_{22} + O(\phi),$$

$$(3.43) \quad vF^v = v^2 v_{11} v_{22} + 1 + O(\phi).$$

Let us simplify the six coefficients of the third-order derivatives on the right-hand side of (3.41). Throughout these calculations, we will frequently use (3.42) and (3.43).

- The coefficient of $v_{11\alpha}^2$ is

$$\begin{aligned}
& \frac{2F^{11}}{v_{11}} - 2 \left(\frac{F^{11}}{v} + vv_{22} \right) \frac{F^{11}}{F^v} + 2v_{11}v_{22} \left(\frac{F^{11}}{F^v} \right)^2 \\
&= \frac{2F^{11}}{v_{11}(vF^v)^2} ((vF^v)^2 - F^{11}v_{11} \cdot vF^v - v^2 v_{11}v_{22} \cdot vF^v + v^2 v_{11}v_{22} \cdot F^{11}v_{11})
\end{aligned}$$

$$\begin{aligned}
&= \frac{2}{(vF^v)^2} \frac{F^{11}}{v_{11}} (F^{22}v_{22} \cdot vF^v - v^2v_{11}v_{22} \cdot F^{22}v_{22} + O(\phi)) \\
&= \frac{2}{(vF^v)^2} \frac{v_{22}}{v_{11}} F^{11}F^{22} + O(\phi).
\end{aligned}$$

- The coefficient of $v_{12\alpha}^2$ is

$$\begin{aligned}
&\frac{2}{v_{11}v_{22}} (F^{11}v_{11} + F^{22}v_{22} - v^2v_{11}v_{22}) - \frac{8(F^{12})^2}{vF^v} + 8v_{11}v_{22} \left(\frac{F^{12}}{F^v} \right)^2 \\
&= \frac{2}{v_{11}v_{22}} - \frac{8(F^{12})^2}{vF^v} + 8v_{11}v_{22} \left(\frac{F^{12}}{F^v} \right)^2 + O(\phi) \\
&= \frac{2}{(vF^v)^2} \frac{1}{v_{11}v_{22}} ((vF^v)^2 - 4v_{11}v_{22}(F^{12})^2 \cdot vF^v + 4v_{11}v_{22}(F^{12})^2 \cdot v^2v_{11}v_{22}) + O(\phi) \\
&= \frac{2}{(vF^v)^2} \frac{1}{v_{11}v_{22}} ((vF^v)^2 - 4v_{11}v_{22}(F^{12})^2) + O(\phi).
\end{aligned}$$

- The coefficient of $v_{22\alpha}^2$ is

$$\begin{aligned}
&\frac{2F^{22}}{v_{22}} - 2 \left(\frac{F^{22}}{v} + vv_{11} \right) \frac{F^{22}}{F^v} + 2v_{11}v_{22} \left(\frac{F^{22}}{F^v} \right)^2 \\
&= \frac{2}{(vF^v)^2} \frac{v_{11}}{v_{22}} F^{11}F^{22} + O(\phi).
\end{aligned}$$

- The coefficient of $2v_{11\alpha}v_{12\alpha}$ is

$$\begin{aligned}
&\frac{2F^{12}}{v_{11}} - 2 \left(\frac{F^{11}}{v} + vv_{22} \right) \frac{F^{12}}{F^v} - \frac{2F^{11}F^{12}}{vF^v} + 4v_{11}v_{22} \frac{F^{11}F^{12}}{(F^v)^2} \\
&= \frac{2}{(vF^v)^2} \frac{F^{12}}{v_{11}} ((vF^v)^2 - 2F^{11}v_{11} \cdot vF^v - v^2v_{11}v_{22} \cdot vF^v + 2F^{11}v_{11} \cdot v^2v_{11}v_{22}) \\
&= \frac{2}{(vF^v)^2} \frac{F^{12}}{v_{11}} (vF^v - 2F^{11}v_{11}) + O(\phi).
\end{aligned}$$

- The coefficient of $2v_{11\alpha}v_{22\alpha}$ is

$$\begin{aligned}
&v^2 - \left(\frac{F^{11}}{v} + vv_{22} \right) \frac{F^{22}}{F^v} - \left(\frac{F^{22}}{v} + vv_{11} \right) \frac{F^{11}}{F^v} + 2v_{11}v_{22} \frac{F^{11}F^{22}}{(F^v)^2} \\
&= v^2 - \frac{2F^{11}F^{22}}{vF^v} - \frac{v}{F^v} (F^{11}v_{11} + F^{22}v_{22}) + 2v_{11}v_{22} \frac{F^{11}F^{22}}{(F^v)^2} \\
&= \frac{2}{(vF^v)^2} F^{11}F^{22} (-vF^v + v^2v_{11}v_{22}) + O(\phi) \\
&= -\frac{2}{(vF^v)^2} F^{11}F^{22} + O(\phi).
\end{aligned}$$

- The coefficient of $2v_{12\alpha}v_{22\alpha}$ is

$$\frac{2F^{12}}{v_{22}} - 2 \left(\frac{F^{22}}{v} + vv_{11} \right) \frac{F^{12}}{F^v} - \frac{2F^{12}F^{22}}{vF^v} + 4v_{11}v_{22} \frac{F^{12}F^{22}}{(F^v)^2}$$

$$= \frac{2}{(vF^v)^2} \frac{F^{12}}{v_{22}} (vF^v - 2v_{22}F^{22}) + O(\phi).$$

Therefore, formula (3.41) is reduced to

$$\begin{aligned} & \frac{(vF^v)^2}{2} \left(\sum_{i,j,k,l \in G} F^{ij,kl} v_{ij\alpha} v_{kl\alpha} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij\alpha} v_{\alpha} + F^{v,v} v_{\alpha}^2 + 2 \sum_{i,j,\beta \in G} \frac{1}{v_{\beta\beta}} F^{ij} v_{\alpha\beta i} v_{\beta\alpha j} \right) \\ &= \frac{v_{22}}{v_{11}} F^{11} F^{22} v_{11\alpha}^2 + \frac{1}{v_{11}v_{22}} ((vF^v)^2 - 4v_{11}v_{22}(F^{12})^2) v_{12\alpha}^2 + \frac{v_{11}}{v_{22}} F^{11} F^{22} v_{22\alpha}^2 \\ &+ 2 \frac{F^{12}}{v_{11}} (vF^v - 2F^{11}v_{11}) v_{11\alpha} v_{12\alpha} - 2F^{11} F^{22} v_{11\alpha} v_{22\alpha} + 2 \frac{F^{12}}{v_{22}} (vF^v - 2F^{22}v_{22}) v_{12\alpha} v_{22\alpha} \\ &+ O\left(\phi + \sum_{\alpha,\beta \in B} |\nabla v_{\alpha\beta}|\right). \end{aligned}$$

Let us define a quadratic form in three variables

$$f(X) = X^T M X,$$

where $X = (v_{11\alpha}, v_{12\alpha}, v_{22\alpha})^T$ and the symmetric matrix

$$M = \begin{pmatrix} \frac{v_{22}}{v_{11}} F^{11} F^{22} & \frac{F^{12}}{v_{11}} (vF^v - 2F^{11}v_{11}) & -F^{11} F^{22} \\ \frac{F^{12}}{v_{11}} (vF^v - 2F^{11}v_{11}) & \frac{1}{v_{11}v_{22}} ((vF^v)^2 - 4v_{11}v_{22}(F^{12})^2) & \frac{F^{12}}{v_{22}} (vF^v - 2F^{22}v_{22}) \\ -F^{11} F^{22} & \frac{F^{12}}{v_{22}} (vF^v - 2F^{22}v_{22}) & \frac{v_{11}}{v_{22}} F^{11} F^{22} \end{pmatrix}.$$

Then, it suffices to establish the positive semidefiniteness of $f(X)$ in the sense that

$$f(X) \geq -C\phi.$$

This is equivalent to saying that M is positive semidefinite modulo ϕ .

By Lemma 2.2, we need to compute the leading principal minors of the matrix M , denoted by $P_1(M)$, $P_2(M)$ and $P_3(M)$.

- $P_1(M)$. Clearly,

$$P_1(M) = \frac{v_{22}}{v_{11}} F^{11} F^{22}.$$

- $P_2(M)$. By (3.42) and (3.43), we have

$$\begin{aligned} P_2(M) &= \frac{F^{11} F^{22}}{v_{11}^2} ((vF^v)^2 - 4v_{11}v_{22}(F^{12})^2) - \frac{(F^{12})^2}{v_{11}^2} (vF^v - 2F^{11}v_{11})^2 \\ &= \frac{(vF^v)^2}{v_{11}^2} (F^{11} F^{22} - (F^{12})^2) + O(\phi). \end{aligned}$$

- $P_3(M)$. By direct computation and (3.42), we have

$$\begin{aligned} P_3(M) = \det(M) &= - \frac{4F^{11} F^{22} (F^{12})^2}{v_{11}v_{22}} (vF^v - F^{11}v_{11} - F^{22}v_{22})^2 \\ &= O(\phi^2). \end{aligned}$$

Since $P_1(M)$ and $P_2(M)$ are bounded below by positive constants, Lemma 2.2 implies that M is positive semidefinite modulo ϕ . We complete the proof of **Claim 1**. $\square$

As a consequence of Theorem 3.7, if $\nabla^2 v(x)$ attains the minimal rank 2 at some point $x_0 \in \Omega$, then $\nabla^2 v$ has constant rank 2 throughout Ω . We defer the proof of this statement to Theorem 5.1 in Section 5.

4. DIFFERENTIAL INEQUALITY: THE CASE OF MINIMAL RANK 3

In this section, we study the case where the real Hessian $\nabla^2 v$ attains its minimal rank 3 at some point $x_0 \in \Omega$, equivalently, K has rank 1 at x_0 . We define

$$\phi(x) = \det(K(x)).$$

As in the previous section, our goal is to prove a differential inequality for ϕ .

Theorem 4.1. *Let $\Omega \subset \mathbb{R}^4$ be a domain and let $v \in C^4(\Omega)$ be a convex solution to equation (2.6). If the Hessian $\nabla^2 v(x)$ attains its minimal rank 3 at some point $x_0 \in \Omega$, then there exist a neighborhood $\mathcal{O}$ of x_0 and a positive constant C (independent of ϕ) such that*

$$(4.1) \quad \sum_{i,j} F^{ij} \phi_{ij} \leq C(\phi + |\nabla \phi|) \quad \text{in } \mathcal{O}.$$

Proof. Following the notation in [10, 34], for two functions $h(x)$ and $k(x)$ defined in $\mathcal{O}$, we say that $h(x) \lesssim k(x)$ if there exists a positive constant C_1 such that

$$h(x) - k(x) \leq C_1(\phi(x) + |\nabla \phi|(x)).$$

We also write $h(x) \sim k(x)$ if $h(x) \lesssim k(x)$ and $k(x) \lesssim h(x)$. Next, we write $h \lesssim k$ if the above inequality holds for any $x \in \mathcal{O}$, with constant C_1 independent of x . Finally, $h \sim k$ means $h \lesssim k$ and $k \lesssim h$. In a similar way, we will also use the notation $h \gtrsim k$ which means $-h \lesssim -k$, namely,

$$-h + k \leq C_2(\phi + |\nabla \phi|) \quad \text{in } \mathcal{O},$$

or equivalently,

$$h - k \geq -C_2(\phi + |\nabla \phi|) \quad \text{in } \mathcal{O}.$$

Thus, $h \sim k$ can also be interpreted as $h \lesssim k$ and $h \gtrsim k$, meaning that there exist positive constants C_3 and C_4 such that

$$-C_3(\phi + |\nabla \phi|) \leq h - k \leq C_4(\phi + |\nabla \phi|) \quad \text{in } \mathcal{O}.$$

We can also write this as

$$h = k + O(\phi + |\nabla \phi|).$$

For each fixed $x \in \mathcal{O}$, we choose the coordinates introduced in Section 2.2 so that $\nabla^2 v(x)$ is diagonal. In the following, all calculations will be carried out at the fixed point x . When we use the relation $\lesssim$ or $\gtrsim$, all constants involved are under control. After relabeling the two complex coordinates if necessary, we may assume that v_{44} is the “bad”

eigenvalue. Thus, in a sufficiently small neighborhood of x_0 , we set $G = \{1, 2, 3\}$ and $B = \{4\}$. Then there exists a positive constant c such that $v_{11}, v_{22}, v_{33} \geq c$. By (3.4), the auxiliary function ϕ can be written as

$$\phi = \frac{1}{16}(\det(A))^{-2} \det(\nabla^2 v),$$

where A is defined in (2.13). Then we have

$$0 \sim 16\phi = (\det(A))^{-2} \sigma_3(G) v_{44},$$

and

$$(4.2) \quad v_{44} \sim 0.$$

Taking the first derivatives of ϕ , by Lemma 3.1, we get

$$\begin{aligned} 0 \sim 16\phi_i &= ((\det(A))^{-2})_i \det(\nabla^2 v) + (\det(A))^{-2} \sum_{p,q} \frac{\partial \det(\nabla^2 v)}{\partial v_{pq}} v_{pqi} \\ &\sim (\det(A))^{-2} \sum_p \sigma_3(\lambda|p) v_{ppi} \sim (\det(A))^{-2} \sigma_3(G) v_{44i}, \quad i = 1, 2, 3, 4. \end{aligned}$$

Consequently,

$$(4.3) \quad v_{44i} \sim 0, \quad i = 1, 2, 3, 4.$$

Using relations (4.2)–(4.3) and Lemma 3.1, we compute the second derivatives of ϕ as

$$\begin{aligned} 16\phi_{ij} &= ((\det(A))^{-2})_{ij} \det(\nabla^2 v) + ((\det(A))^{-2})_i \sum_{p,q} \frac{\partial \det(\nabla^2 v)}{\partial v_{pq}} v_{pqj} \\ &\quad + ((\det(A))^{-2})_j \sum_{p,q} \frac{\partial \det(\nabla^2 v)}{\partial v_{pq}} v_{pqi} + (\det(A))^{-2} \sum_{p,q} \frac{\partial \det(\nabla^2 v)}{\partial v_{pq}} v_{pqij} \\ &\quad + (\det(A))^{-2} \sum_{p,q,r,s} \frac{\partial^2 \det(\nabla^2 v)}{\partial v_{pq} \partial v_{rs}} v_{pqi} v_{rsj} \\ &= ((\det(A))^{-2})_{ij} \sigma_3(G) v_{44} + ((\det(A))^{-2})_i \sum_p \sigma_3(\lambda|p) v_{ppj} \\ &\quad + ((\det(A))^{-2})_j \sum_p \sigma_3(\lambda|p) v_{ppi} + (\det(A))^{-2} \sum_p \sigma_3(\lambda|p) v_{ppij} \\ &\quad + (\det(A))^{-2} \sum_{p \neq r} \sigma_2(\lambda|pr) v_{ppi} v_{rrj} - (\det(A))^{-2} \sum_{p \neq r} \sigma_2(\lambda|pr) v_{pri} v_{rppj} \\ &\sim (\det(A))^{-2} \sigma_3(G) v_{44ij} - 2(\det(A))^{-2} \sigma_3(G) \sum_{p \in G} \frac{1}{v_{pp}} v_{4pi} v_{4pj}. \end{aligned}$$

It follows that

$$(4.4) \quad -\frac{16(\det(A))^2}{\sigma_3(G)} \sum_{i,j} F^{ij} \phi_{ij} \sim -\sum_{i,j} F^{ij} v_{44ij} + 2 \sum_{p \in G} \sum_{i,j} \frac{1}{v_{pp}} F^{ij} v_{4pi} v_{4pj}.$$

To handle the fourth-order derivative terms in (4.4), we first take the derivative of the equation $F(\nabla^2 v, \nabla v, v) = 0$ with respect to the variable x_4 , resulting in

$$(4.5) \quad 0 = \sum_{i,j} F^{ij} v_{ij4} + \sum_k F^{v_k} v_{k4} + F^v v_4.$$

Then we take the second derivative, which gives

$$(4.6) \quad \begin{aligned} 0 = & \sum_{i,j,k,l} F^{ij,kl} v_{ij4} v_{kl4} + \sum_{i,j,k} F^{ij,v_k} v_{ij4} v_{k4} + \sum_{i,j} F^{ij,v} v_{ij4} v_4 + \sum_{i,j} F^{ij} v_{ij44} \\ & + \sum_{k,i,j} F^{v_k,ij} v_{k4} v_{ij4} + \sum_{k,l} F^{v_k,v_l} v_{k4} v_{l4} + \sum_k F^{v_k,v} v_{k4} v_4 + \sum_k F^{v_k} v_{k44} \\ & + \sum_{i,j} F^{v,ij} v_4 v_{ij4} + \sum_k F^{v,v_k} v_4 v_{k4} + F^{v,v} v_4^2 + F^v v_{44}. \end{aligned}$$

By (4.2) and (4.3), equations (4.5) and (4.6) can be simplified as

$$(4.7) \quad 0 \sim \sum_{i,j \in G} F^{ij} v_{ij4} + F^v v_4$$

and

$$(4.8) \quad - \sum_{i,j} F^{ij} v_{ij44} \sim \sum_{i,j,k,l \in G} F^{ij,kl} v_{ij4} v_{kl4} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij4} v_4 + F^{v,v} v_4^2.$$

Substituting (4.8) into (4.4), we have

$$(4.9) \quad \begin{aligned} - \frac{16(\det(A))^2}{\sigma_3(G)} \sum_{i,j} F^{ij} \phi_{ij} \sim & \sum_{i,j,k,l \in G} F^{ij,kl} v_{ij4} v_{kl4} + 2 \sum_{i,j \in G} F^{ij,v} v_{ij4} v_4 + F^{v,v} v_4^2 \\ & + 2 \sum_{p \in G} \sum_{i,j} \frac{1}{v_{pp}} F^{ij} v_{4pi} v_{4pj}. \end{aligned}$$

Next, we carefully analyze each term on the right-hand side of (4.9). Recall that

$$\begin{aligned} F(\nabla^2 v, \nabla v, v) = & v^2 [(v_{11} + v_{33})(v_{22} + v_{44}) - (v_{12} + v_{34})(v_{21} + v_{43}) - (v_{14} - v_{32})(v_{41} - v_{23})] \\ & + v [(v_2^2 + v_4^2)(v_{11} + v_{33}) + (v_1^2 + v_3^2)(v_{22} + v_{44}) \\ & - (v_1 v_2 + v_3 v_4)(v_{12} + v_{34} + v_{21} + v_{43}) \\ & - (v_1 v_4 - v_2 v_3)(v_{14} - v_{32} + v_{41} - v_{23})] - 1. \end{aligned}$$

Direct calculation implies

$$\begin{aligned} F^{11,22} = F^{22,11} = F^{22,33} = F^{33,22} = v^2, \\ F^{12,21} = F^{21,12} = F^{23,32} = F^{32,23} = -v^2. \end{aligned}$$

By collecting the terms in equation (4.9), we obtain

$$\begin{aligned}
(4.10) \quad & -\frac{16(\det(A))^2}{\sigma_3(G)} \sum_{i,j} F^{ij} \phi_{ij} \sim \frac{2F^{11}}{v_{11}} v_{114}^2 + \frac{2F^{22}}{v_{22}} v_{224}^2 + \frac{2F^{33}}{v_{33}} v_{334}^2 + \left(\frac{2F^{33}}{v_{11}} + \frac{2F^{11}}{v_{33}} \right) v_{134}^2 \\
& + \left(-2v^2 + \frac{2F^{22}}{v_{11}} + \frac{2F^{11}}{v_{22}} \right) v_{124}^2 + \left(-2v^2 + \frac{2F^{33}}{v_{22}} + \frac{2F^{22}}{v_{33}} \right) v_{234}^2 \\
& + 2v^2 v_{114} v_{224} + \frac{4F^{12}}{v_{11}} v_{114} v_{124} + 2v^2 v_{224} v_{334} + \frac{4F^{12}}{v_{22}} v_{224} v_{124} \\
& + \frac{4F^{23}}{v_{22}} v_{224} v_{234} + \frac{4F^{23}}{v_{33}} v_{334} v_{234} + \frac{4F^{23}}{v_{11}} v_{134} v_{124} + \frac{4F^{12}}{v_{33}} v_{134} v_{234} \\
& + 2F^{11,v} v_{114} v_4 + 2F^{22,v} v_{224} v_4 + 2F^{33,v} v_{334} v_4 + 4F^{12,v} v_{124} v_4 \\
& + 4F^{23,v} v_{234} v_4 + F^{v,v} v_4^2.
\end{aligned}$$

It is easy to see that

$$(4.11) \quad vF^v = v^2(v_{11} + v_{33})(v_{22} + v_{44}) + 1.$$

This implies $F^v \neq 0$. Since $F^{13} = 0$, by (4.7), we have

$$(4.12) \quad v_4 \sim -\frac{F^{11}}{F^v} v_{114} - \frac{F^{22}}{F^v} v_{224} - \frac{F^{33}}{F^v} v_{334} - \frac{2F^{12}}{F^v} v_{124} - \frac{2F^{23}}{F^v} v_{234}.$$

Putting (4.12) into (4.10), we obtain a quadratic form in terms of v_{114} , v_{224} , v_{334} , v_{134} , v_{124} and v_{234} , given by

$$\begin{aligned}
& -\frac{16(\det(A))^2}{\sigma_3(G)} \sum_{i,j} F^{ij} \phi_{ij} \sim \left(\frac{2F^{11}}{v_{11}} - \frac{2F^{11}F^{11,v}}{F^v} + \left(\frac{F^{11}}{F^v} \right)^2 F^{v,v} \right) v_{114}^2 \\
& + \left(\frac{2F^{22}}{v_{22}} - \frac{2F^{22}F^{22,v}}{F^v} + \left(\frac{F^{22}}{F^v} \right)^2 F^{v,v} \right) v_{224}^2 \\
& + \left(\frac{2F^{33}}{v_{33}} - \frac{2F^{33}F^{33,v}}{F^v} + \left(\frac{F^{33}}{F^v} \right)^2 F^{v,v} \right) v_{334}^2 \\
& + \left(\frac{2F^{33}}{v_{11}} + \frac{2F^{11}}{v_{33}} \right) v_{134}^2 \\
& + \left(-2v^2 + \frac{2F^{22}}{v_{11}} + \frac{2F^{11}}{v_{22}} - \frac{8F^{12}F^{12,v}}{F^v} + 4 \left(\frac{F^{12}}{F^v} \right)^2 F^{v,v} \right) v_{124}^2 \\
& + \left(-2v^2 + \frac{2F^{33}}{v_{22}} + \frac{2F^{22}}{v_{33}} - \frac{8F^{23}F^{23,v}}{F^v} + 4 \left(\frac{F^{23}}{F^v} \right)^2 F^{v,v} \right) v_{234}^2 \\
& + 2 \left(v^2 - \frac{F^{22}F^{11,v}}{F^v} - \frac{F^{11}F^{22,v}}{F^v} + \frac{F^{11}F^{22}}{(F^v)^2} F^{v,v} \right) v_{114} v_{224}
\end{aligned}$$

$$\begin{aligned}
& + 2 \left(-\frac{F^{33}F^{11,v}}{F^v} - \frac{F^{11}F^{33,v}}{F^v} + \frac{F^{11}F^{33}}{(F^v)^2}F^{v,v} \right) v_{114}v_{334} \\
& + 2 \left(\frac{2F^{12}}{v_{11}} - \frac{2F^{12}F^{11,v}}{F^v} - \frac{2F^{11}F^{12,v}}{F^v} + \frac{2F^{11}F^{12}}{(F^v)^2}F^{v,v} \right) v_{114}v_{124} \\
& + 2 \left(-\frac{2F^{23}F^{11,v}}{F^v} - \frac{2F^{11}F^{23,v}}{F^v} + \frac{2F^{11}F^{23}}{(F^v)^2}F^{v,v} \right) v_{114}v_{234} \\
& + 2 \left(v^2 - \frac{F^{33}F^{22,v}}{F^v} - \frac{F^{22}F^{33,v}}{F^v} + \frac{F^{22}F^{33}}{(F^v)^2}F^{v,v} \right) v_{224}v_{334} \\
& + 2 \left(\frac{2F^{12}}{v_{22}} - \frac{2F^{12}F^{22,v}}{F^v} - \frac{2F^{22}F^{12,v}}{F^v} + \frac{2F^{12}F^{22}}{(F^v)^2}F^{v,v} \right) v_{224}v_{124} \\
& + 2 \left(\frac{2F^{23}}{v_{22}} - \frac{2F^{23}F^{22,v}}{F^v} - \frac{2F^{22}F^{23,v}}{F^v} + \frac{2F^{22}F^{23}}{(F^v)^2}F^{v,v} \right) v_{224}v_{234} \\
& + 2 \left(-\frac{2F^{12}F^{33,v}}{F^v} - \frac{2F^{33}F^{12,v}}{F^v} + \frac{2F^{12}F^{33}}{(F^v)^2}F^{v,v} \right) v_{334}v_{124} \\
& + 2 \left(\frac{2F^{23}}{v_{33}} - \frac{2F^{23}F^{33,v}}{F^v} - \frac{2F^{33}F^{23,v}}{F^v} + \frac{2F^{23}F^{33}}{(F^v)^2}F^{v,v} \right) v_{334}v_{234} \\
& + 2 \left(\frac{2F^{23}}{v_{11}} \right) v_{134}v_{124} + 2 \left(\frac{2F^{12}}{v_{33}} \right) v_{134}v_{234} \\
& + 2 \left(-\frac{4F^{23}F^{12,v}}{F^v} - \frac{4F^{12}F^{23,v}}{F^v} + \frac{4F^{12}F^{23}}{(F^v)^2}F^{v,v} \right) v_{124}v_{234}.
\end{aligned}$$

It suffices to verify that the above quadratic form in the variables v_{114} , v_{224} , v_{334} , v_{134} , v_{124} and v_{234} is positive semidefinite modulo ϕ . To achieve this, we must carefully arrange the coefficients.

Before arranging these coefficients, we recall the equation $F(\nabla^2 v, \nabla v, v) = 0$. It is equivalent to

$$(4.13) \quad 0 \sim v^2(v_{11} + v_{33})v_{22} + v(v_2^2 + v_4^2)(v_{11} + v_{33}) + v(v_1^2 + v_3^2)v_{22} - 1.$$

At the point under consideration, recall from (2.8) that the positive definite matrix (F^{ij}) is given by

$$(4.14) \quad (F^{ij}) = \begin{pmatrix} F^{11} & F^{12} & 0 & -F^{23} \\ F^{12} & F^{22} & F^{23} & 0 \\ 0 & F^{23} & F^{11} & F^{12} \\ -F^{23} & 0 & F^{12} & F^{22} \end{pmatrix},$$

where

$$(4.15) \quad \begin{aligned} F^{11} & \sim v^2v_{22} + v(v_2^2 + v_4^2), & F^{12} & = -v(v_1v_2 + v_3v_4), \\ F^{22} & = v^2(v_{11} + v_{33}) + v(v_1^2 + v_3^2), & F^{23} & = v(v_1v_4 - v_2v_3). \end{aligned}$$

Since $v_{44} \sim 0$, by (4.11), we have

$$(4.16) \quad vF^v \sim v^2(v_{11} + v_{33})v_{22} + 1.$$

Combining this with (4.14)–(4.15), we also have

$$(4.17) \quad vF^v \sim F^{11}v_{11} + F^{22}v_{22} + F^{33}v_{33}.$$

Furthermore, a simple computation gives

$$(4.18) \quad \begin{aligned} F^{11,v} &\sim 2vv_{22} + (v_2^2 + v_4^2) \sim F^{33,v}, \\ F^{22,v} &\sim 2v(v_{11} + v_{33}) + (v_1^2 + v_3^2), \\ F^{12,v} &\sim -(v_1v_2 + v_3v_4), \quad F^{23,v} \sim v_1v_4 - v_2v_3, \\ F^{v,v} &\sim 2(v_{11} + v_{33})v_{22}. \end{aligned}$$

We now begin to simplify the coefficients of all terms in the final quadratic form. Throughout this process, we will repeatedly use (4.14)–(4.18).

- The coefficient of v_{11}^2 is

$$\begin{aligned} & \frac{2F^{11}}{v_{11}} - \frac{2F^{11}F^{11,v}}{F^v} + \left(\frac{F^{11}}{F^v} \right)^2 F^{v,v} \\ &= \frac{1}{(vF^v)^2} \frac{F^{11}}{v_{11}} (2(vF^v)^2 - 2v_{11} \cdot vF^{11,v} \cdot vF^v + F^{11}v_{11} \cdot v^2F^{v,v}) \\ &\sim \frac{1}{(vF^v)^2} \frac{F^{11}}{v_{11}} (2(v^2(v_{11} + v_{33})v_{22} + 1) \cdot vF^v - 2v_{11} \cdot (F^{11} + v^2v_{22}) \cdot vF^v \\ &\quad + F^{11}v_{11} \cdot 2v^2(v_{11} + v_{33})v_{22}) \\ &= \frac{2}{(vF^v)^2} \frac{F^{11}}{v_{11}} (v^2v_{22}v_{33} \cdot vF^v + vF^v - F^{11}v_{11} \cdot vF^v + F^{11}v_{11} \cdot v^2(v_{11} + v_{33})v_{22}) \\ &\sim \frac{2}{(vF^v)^2} \frac{F^{11}}{v_{11}} (v^2v_{22}v_{33} \cdot vF^v + vF^v - F^{11}v_{11}) \\ &\sim \frac{2}{(vF^v)^2} \frac{F^{11}}{v_{11}} (v^2v_{22}v_{33} \cdot vF^v + F^{22}v_{22} + F^{33}v_{33}). \end{aligned}$$

- The coefficient of v_{22}^2 is

$$\begin{aligned} & \frac{2F^{22}}{v_{22}} - \frac{2F^{22}F^{22,v}}{F^v} + \left(\frac{F^{22}}{F^v} \right)^2 F^{v,v} \\ &= \frac{1}{(vF^v)^2} \frac{F^{22}}{v_{22}} (2(vF^v)^2 - 2v_{22} \cdot vF^{22,v} \cdot vF^v + F^{22}v_{22} \cdot v^2F^{v,v}) \\ &\sim \frac{1}{(vF^v)^2} \frac{F^{22}}{v_{22}} (2(v^2(v_{11} + v_{33})v_{22} + 1) \cdot vF^v - 2v_{22} \cdot (F^{22} + v^2(v_{11} + v_{33})) \cdot vF^v \\ &\quad + F^{22}v_{22} \cdot 2v^2(v_{11} + v_{33})v_{22}) \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{(vF^v)^2} \frac{F^{22}}{v_{22}} (vF^v - F^{22}v_{22} \cdot vF^v + F^{22}v_{22} \cdot v^2(v_{11} + v_{33})v_{22}) \\
&\sim \frac{2}{(vF^v)^2} \frac{F^{22}}{v_{22}} (vF^v - F^{22}v_{22}) \\
&\sim \frac{2}{(vF^v)^2} \frac{F^{22}}{v_{22}} (F^{11}v_{11} + F^{33}v_{33}).
\end{aligned}$$

- The coefficient of v_{334}^2 is

$$\begin{aligned}
&\frac{2F^{33}}{v_{33}} - \frac{2F^{33}F^{33,v}}{F^v} + \left(\frac{F^{33}}{F^v}\right)^2 F^{v,v} \\
&\sim \frac{2}{(vF^v)^2} \frac{F^{33}}{v_{33}} (v^2v_{11}v_{22} \cdot vF^v + F^{11}v_{11} + F^{22}v_{22}),
\end{aligned}$$

where we exploited the symmetry between indices 1 and 3.

- The coefficient of v_{134}^2 is

$$\frac{2F^{33}}{v_{11}} + \frac{2F^{11}}{v_{33}}.$$

- The coefficient of v_{124}^2 is

$$\begin{aligned}
&-2v^2 + \frac{2F^{22}}{v_{11}} + \frac{2F^{11}}{v_{22}} - \frac{8F^{12}F^{12,v}}{F^v} + 4\left(\frac{F^{12}}{F^v}\right)^2 F^{v,v} \\
&= \frac{2}{(vF^v)^2} \left(\left(-v^2 + \frac{F^{22}}{v_{11}} + \frac{F^{11}}{v_{22}} \right) \cdot (vF^v)^2 - 4F^{12} \cdot vF^{12,v} \cdot vF^v + 2(F^{12})^2 \cdot v^2F^{v,v} \right) \\
&\sim \frac{2}{(vF^v)^2} \left(\frac{-v^2v_{11}v_{22} + F^{22}v_{22} + F^{11}v_{11}}{v_{11}v_{22}} (vF^v)^2 \right. \\
&\quad \left. - 4(F^{12})^2 \left(v^2(v_{11} + v_{33})v_{22} + 1 \right) + 4(F^{12})^2 \cdot v^2(v_{11} + v_{33})v_{22} \right) \\
&\sim \frac{2}{(vF^v)^2} \left(\frac{v^2v_{22}v_{33} + 1 - F^{33}v_{33}}{v_{11}v_{22}} (vF^v)^2 - 4(F^{12})^2 \right).
\end{aligned}$$

- The coefficient of v_{234}^2 is

$$\begin{aligned}
&-2v^2 + \frac{2F^{33}}{v_{22}} + \frac{2F^{22}}{v_{33}} - \frac{8F^{23}F^{23,v}}{F^v} + 4\left(\frac{F^{23}}{F^v}\right)^2 F^{v,v} \\
&\sim \frac{2}{(vF^v)^2} \left(\frac{v^2v_{11}v_{22} + 1 - F^{11}v_{11}}{v_{22}v_{33}} (vF^v)^2 - 4(F^{23})^2 \right).
\end{aligned}$$

- The coefficient of $2v_{114}v_{224}$ is

$$v^2 - \frac{F^{22}F^{11,v}}{F^v} - \frac{F^{11}F^{22,v}}{F^v} + \frac{F^{11}F^{22}}{(F^v)^2} F^{v,v}$$

$$\begin{aligned}
&= \frac{1}{(vF^v)^2} (v^2 \cdot (vF^v)^2 - F^{22} \cdot vF^{11,v} \cdot vF^v - F^{11} \cdot vF^{22,v} \cdot vF^v + F^{11}F^{22} \cdot v^2F^{v,v}) \\
&\sim \frac{1}{(vF^v)^2} (v^2 \cdot (vF^v)^2 - F^{22} \cdot (F^{11} + v^2v_{22}) \cdot vF^v) - F^{11} \cdot (F^{22} + v^2(v_{11} + v_{33})) \cdot vF^v \\
&\quad + F^{11}F^{22} \cdot 2v^2(v_{11} + v_{33})v_{22}) \\
&= \frac{1}{(vF^v)^2} (v^2 \cdot (vF^v)^2 - v^2 \cdot (F^{22}v_{22} + F^{11}v_{11} + F^{11}v_{33}) \cdot vF^v - 2F^{11}F^{22} \\
&\quad + 2F^{11}F^{22} \cdot v^2(v_{11} + v_{33})v_{22}) \\
&\sim -\frac{2}{(vF^v)^2} F^{11}F^{22}.
\end{aligned}$$

- The coefficient of $2v_{114}v_{334}$ is

$$\begin{aligned}
&-\frac{F^{33}F^{11,v}}{F^v} - \frac{F^{11}F^{33,v}}{F^v} + \frac{F^{11}F^{33}}{(F^v)^2}F^{v,v} \\
&= -\frac{1}{(vF^v)^2} (F^{33} \cdot vF^{11,v} \cdot vF^v + F^{11} \cdot vF^{33,v} \cdot vF^v - F^{11}F^{33} \cdot v^2F^{v,v}) \\
&\sim -\frac{1}{(vF^v)^2} (F^{33} \cdot (F^{11} + v^2v_{22}) \cdot vF^v + F^{11} \cdot (F^{33} + v^2v_{22}) \cdot vF^v \\
&\quad - F^{11}F^{33} \cdot 2v^2(v_{11} + v_{33})v_{22}) \\
&= -\frac{1}{(vF^v)^2} (2F^{11}F^{33} - 2F^{11}F^{33} \cdot v^2(v_{11} + v_{33})v_{22} + (F^{33} + F^{11}) \cdot v^2v_{22} \cdot vF^v) \\
&\sim -\frac{2F^{11}}{(vF^v)^2} (F^{11} + v^2v_{22} \cdot vF^v).
\end{aligned}$$

- The coefficient of $2v_{114}v_{124}$ is

$$\begin{aligned}
&\frac{2F^{12}}{v_{11}} - \frac{2F^{12}F^{11,v}}{F^v} - \frac{2F^{11}F^{12,v}}{F^v} + \frac{2F^{11}F^{12}}{(F^v)^2}F^{v,v} \\
&= \frac{2}{(vF^v)^2} \left(\frac{F^{12}}{v_{11}}(vF^v)^2 - F^{12} \cdot vF^{11,v} \cdot vF^v - F^{11} \cdot vF^{12,v} \cdot vF^v + F^{11}F^{12} \cdot v^2F^{v,v} \right) \\
&\sim \frac{2}{(vF^v)^2} \left(\frac{F^{12}}{v_{11}} (v^2(v_{11} + v_{33})v_{22} + 1) \cdot vF^v - F^{12} \cdot (F^{11} + v^2v_{22}) \cdot vF^v \right. \\
&\quad \left. - F^{11}F^{12} \cdot vF^v + F^{11}F^{12} \cdot 2v^2(v_{11} + v_{33})v_{22} \right) \\
&= \frac{2}{(vF^v)^2} \left(\frac{F^{12}}{v_{11}} (v^2v_{22}v_{33} + 1) \cdot vF^v - 2F^{11}F^{12}(vF^v - v^2(v_{11} + v_{33})v_{22}) \right) \\
&\sim \frac{2}{(vF^v)^2} \frac{F^{12}}{v_{11}} (v^2v_{22}v_{33} \cdot vF^v + vF^v - 2F^{11}v_{11}).
\end{aligned}$$

- The coefficient of $2v_{114}v_{234}$ is

$$\begin{aligned}
& -\frac{2F^{23}F^{11,v}}{F^v} - \frac{2F^{11}F^{23,v}}{F^v} + \frac{2F^{11}F^{23}}{(F^v)^2}F^{v,v} \\
& = -\frac{2}{(vF^v)^2}(F^{23} \cdot vF^{11,v} \cdot vF^v + F^{11} \cdot vF^{23,v} \cdot vF^v - F^{11}F^{23} \cdot v^2F^{v,v}) \\
& \sim -\frac{2}{(vF^v)^2}(F^{23} \cdot (F^{11} + v^2v_{22}) \cdot vF^v + F^{11}F^{23} \cdot vF^v - F^{11}F^{23} \cdot 2v^2(v_{11} + v_{33})v_{22}) \\
& = -\frac{2}{(vF^v)^2}(F^{23} \cdot v^2v_{22} \cdot vF^v + 2F^{11}F^{23} \cdot (vF^v - v^2(v_{11} + v_{33})v_{22})) \\
& \sim -\frac{2F^{23}}{(vF^v)^2}(v^2v_{22} \cdot vF^v + 2F^{11}).
\end{aligned}$$

- The coefficient of $2v_{224}v_{334}$ is

$$v^2 - \frac{F^{33}F^{22,v}}{F^v} - \frac{F^{22}F^{33,v}}{F^v} + \frac{F^{22}F^{33}}{(F^v)^2}F^{v,v} \sim -\frac{2}{(vF^v)^2}F^{22}F^{33}.$$

- The coefficient of $2v_{224}v_{124}$ is

$$\begin{aligned}
& \frac{2F^{12}}{v_{22}} - \frac{2F^{12}F^{22,v}}{F^v} - \frac{2F^{22}F^{12,v}}{F^v} + \frac{2F^{12}F^{22}}{(F^v)^2}F^{v,v} \\
& = \frac{2}{(vF^v)^2} \left(\frac{F^{12}}{v_{22}}(vF^v)^2 - F^{12} \cdot vF^{22,v} \cdot vF^v - F^{22} \cdot vF^{12,v} \cdot vF^v + F^{12}F^{22} \cdot v^2F^{v,v} \right) \\
& \sim \frac{2}{(vF^v)^2} \left(\frac{F^{12}}{v_{22}}(vF^v)^2 - F^{12} \cdot (F^{22} + v^2(v_{11} + v_{33})) \cdot vF^v - F^{12}F^{22} \cdot vF^v \right. \\
& \quad \left. + F^{12}F^{22} \cdot 2v^2(v_{11} + v_{33})v_{22} \right) \\
& \sim \frac{2}{(vF^v)^2} \frac{F^{12}}{v_{22}} ((vF^v)^2 - v^2(v_{11} + v_{33})v_{22} \cdot vF^v - 2F^{22}v_{22} \cdot (vF^v - v^2(v_{11} + v_{33})v_{22})) \\
& \sim \frac{2}{(vF^v)^2} \frac{F^{12}}{v_{22}} (vF^v - 2F^{22}v_{22}).
\end{aligned}$$

- The coefficient of $2v_{224}v_{234}$ is

$$\begin{aligned}
& \frac{2F^{23}}{v_{22}} - \frac{2F^{23}F^{22,v}}{F^v} - \frac{2F^{22}F^{23,v}}{F^v} + \frac{2F^{22}F^{23}}{(F^v)^2}F^{v,v} \\
& \sim \frac{2}{(vF^v)^2} \frac{F^{23}}{v_{22}} (vF^v - 2F^{22}v_{22}).
\end{aligned}$$

- The coefficient of $2v_{334}v_{124}$ is

$$-\frac{2F^{12}F^{33,v}}{F^v} - \frac{2F^{33}F^{12,v}}{F^v} + \frac{2F^{12}F^{33}}{(F^v)^2}F^{v,v}$$

$$\sim -\frac{2F^{12}}{(vF^v)^2}(v^2v_{22} \cdot vF^v + 2F^{33}).$$

- The coefficient of $2v_{334}v_{234}$ is

$$\begin{aligned} & \frac{2F^{23}}{v_{33}} - \frac{2F^{23}F^{33,v}}{F^v} - \frac{2F^{33}F^{23,v}}{F^v} + \frac{2F^{23}F^{33}}{(F^v)^2}F^{v,v} \\ & \sim \frac{2}{(vF^v)^2} \frac{F^{23}}{v_{33}}(v^2v_{11}v_{22} \cdot vF^v + vF^v - 2F^{33}v_{33}). \end{aligned}$$

- The coefficient of $2v_{124}v_{234}$ is

$$\begin{aligned} & -\frac{4F^{23}F^{12,v}}{F^v} - \frac{4F^{12}F^{23,v}}{F^v} + \frac{4F^{12}F^{23}}{(F^v)^2}F^{v,v} \\ & = -\frac{4}{(vF^v)^2}(F^{23} \cdot vF^{12,v} \cdot vF^v + F^{12} \cdot vF^{23,v} \cdot vF^v - F^{12}F^{23} \cdot v^2F^{v,v}) \\ & \sim -\frac{8F^{12}F^{23}}{(vF^v)^2}(vF^v - v^2(v_{11} + v_{33})v_{22}) \\ & \sim -\frac{8F^{12}F^{23}}{(vF^v)^2}. \end{aligned}$$

- The coefficients of $2v_{114}v_{134}$, $2v_{224}v_{134}$, $2v_{334}v_{134}$, $2v_{134}v_{124}$ and $2v_{134}v_{234}$ are

$$0, \quad 0, \quad 0, \quad \frac{2F^{23}}{v_{11}} \quad \text{and} \quad \frac{2F^{12}}{v_{33}},$$

respectively.

Therefore, we obtain

$$(4.19) \quad -\frac{16(\det(A))^2(vF^v)^2}{2\sigma_3(G)} \sum_{i,j} F^{ij}\phi_{ij} \sim f(X),$$

where $f(X) = X^T M X$ is a quadratic form with $X = (v_{114}, v_{224}, v_{334}, v_{134}, v_{124}, v_{234})^T$ as the associated vector and M as the symmetric matrix defined below.

	v_{114}	v_{224}	v_{334}	v_{134}	v_{124}	v_{234}
v_{114}	$\frac{F^{11}}{v_{11}}(v^2 v_{22} v_{33} \cdot v F^v + F^{22} v_{22} + F^{33} v_{33})$	$-F^{11} F^{22}$	$-F^{11}(v^2 v_{22} \cdot v F^v + F^{11})$	0	$\frac{F^{12}}{v_{11}}(v^2 v_{22} v_{33} \cdot v F^v + v F^v - 2F^{11} v_{11})$	$-F^{23}(v^2 v_{22} \cdot v F^v + 2F^{11})$
v_{224}	$-F^{11} F^{22}$	$\frac{F^{22}}{v_{22}}(F^{11} v_{11} + F^{33} v_{33})$	$-F^{22} F^{33}$	0	$\frac{F^{12}}{v_{22}}(v F^v - 2F^{22} v_{22})$	$\frac{F^{23}}{v_{22}}(v F^v - 2F^{22} v_{22})$
v_{334}	$-F^{11}(v^2 v_{22} \cdot v F^v + F^{11})$	$-F^{22} F^{33}$	$\frac{F^{33}}{v_{33}}(v^2 v_{11} v_{22} \cdot v F^v + F^{11} v_{11} + F^{22} v_{22})$	0	$-F^{12}(v^2 v_{22} \cdot v F^v + 2F^{33})$	$\frac{F^{23}}{v_{33}}(v^2 v_{11} v_{22} \cdot v F^v + v F^v - 2F^{33} v_{33})$
v_{134}	0	0	0	$\left(\frac{F^{33}}{v_{11}} + \frac{F^{11}}{v_{33}}\right) \cdot (v F^v)^2$	$\frac{F^{23}}{v_{11}}(v F^v)^2$	$\frac{F^{12}}{v_{33}}(v F^v)^2$
v_{124}	$\frac{F^{12}}{v_{11}}(v^2 v_{22} v_{33} \cdot v F^v + v F^v - 2F^{11} v_{11})$	$\frac{F^{12}}{v_{22}}(v F^v - 2F^{22} v_{22})$	$-F^{12}(v^2 v_{22} \cdot v F^v + 2F^{33})$	$\frac{F^{23}}{v_{11}}(v F^v)^2$	$\frac{v^2 v_{22} v_{33} + 1 - F^{33} v_{33}}{(v F^v)^2 - 4(F^{12})^2}$	$-4F^{12} F^{23}$
v_{234}	$-F^{23}(v^2 v_{22} \cdot v F^v + 2F^{11})$	$\frac{F^{23}}{v_{22}}(v F^v - 2F^{22} v_{22})$	$\frac{F^{23}}{v_{33}}(v^2 v_{11} v_{22} \cdot v F^v + v F^v - 2F^{33} v_{33})$	$\frac{F^{12}}{v_{33}}(v F^v)^2$	$-4F^{12} F^{23}$	$\frac{v^2 v_{11} v_{22} + 1 - F^{11} v_{11}}{(v F^v)^2 - 4(F^{23})^2}$

Let us verify that the matrix M is positive semidefinite modulo ϕ , i.e., the corresponding quadratic form $f(X)$ satisfies $f(X) \gtrsim 0$. This is highly technical and challenging.

First, we perform congruence transformations on the matrix M , which preserve the positive semidefiniteness of M , to convert it into a block diagonal form. The specific transformations are as follows.

- (1) The first column is multiplied by $-\frac{F^{12}}{F^{11}}$ and added to the fifth column, while the first row is multiplied by $-\frac{F^{12}}{F^{11}}$ and added to the fifth row. Then, the second column is multiplied by $-\frac{F^{12}}{F^{22}}$ and added to the fifth column, while the second row is multiplied by $-\frac{F^{12}}{F^{22}}$ and added to the fifth row.
- (2) The third column is multiplied by $-\frac{F^{23}}{F^{11}}$ and added to the sixth column, while the third row is multiplied by $-\frac{F^{23}}{F^{11}}$ and added to the sixth row. Then, the second column is multiplied by $-\frac{F^{23}}{F^{22}}$ and added to the sixth column, while the second row is multiplied by $-\frac{F^{23}}{F^{22}}$ and added to the sixth row.

These operations yield the following block diagonal matrix.

	v_{114}	v_{224}	v_{334}	v_{134}	v_{124}	v_{234}
v_{114}	$\frac{F^{11}}{v_{11}}(v^2 v_{22} v_{33} \cdot v F^v + F^{22} v_{22} + F^{33} v_{33})$	$-F^{11} F^{22}$	$-F^{11}(v^2 v_{22} \cdot v F^v + F^{11})$	0	0	0
v_{224}	$-F^{11} F^{22}$	$\frac{F^{22}}{v_{22}}(F^{11} v_{11} + F^{33} v_{33})$	$-F^{22} F^{33}$	0	0	0
v_{334}	$-F^{11}(v^2 v_{22} \cdot v F^v + F^{11})$	$-F^{22} F^{33}$	$\frac{F^{33}}{v_{33}}(v^2 v_{11} v_{22} \cdot v F^v + F^{11} v_{11} + F^{22} v_{22})$	0	0	0
v_{134}	0	0	0	$\left(\frac{F^{33}}{v_{11}} + \frac{F^{11}}{v_{33}}\right) \cdot (v F^v)^2$	$\frac{F^{23}}{v_{11}}(v F^v)^2$	$\frac{F^{12}}{v_{33}}(v F^v)^2$
v_{124}	0	0	0	$\frac{F^{23}}{v_{11}}(v F^v)^2$	b_{11}	b_{12}
v_{234}	0	0	0	$\frac{F^{12}}{v_{33}}(v F^v)^2$	b_{21}	b_{22}

Here

$$b_{11} = \left(\frac{F^{11}}{v_{22}} + \frac{F^{22}}{v_{11}} - v^2 \right) \cdot (v F^v)^2 - \left(\frac{v^2 v_{22} v_{33} + 1}{F^{11} v_{11}} + \frac{1}{F^{22} v_{22}} \right) (F^{12})^2 \cdot v F^v,$$

$$b_{12} = b_{21} = F^{12} F^{23} \left(\frac{v^2 v_{22}}{F^{11}} - \frac{1}{F^{22} v_{22}} \right) \cdot v F^v,$$

and

$$b_{22} = \left(\frac{F^{22}}{v_{33}} + \frac{F^{33}}{v_{22}} - v^2 \right) \cdot (v F^v)^2 - \left(\frac{1}{F^{22} v_{22}} + \frac{v^2 v_{11} v_{22} + 1}{F^{33} v_{33}} \right) (F^{23})^2 \cdot v F^v.$$

Claim 2. The matrix

$$M_1 = \begin{pmatrix} \frac{F^{11}}{v_{11}}(v^2 v_{22} v_{33} \cdot v F^v + F^{22} v_{22} + F^{33} v_{33}) & -F^{11} F^{22} & -F^{11}(v^2 v_{22} \cdot v F^v + F^{11}) \\ -F^{11} F^{22} & \frac{F^{22}}{v_{22}}(F^{11} v_{11} + F^{33} v_{33}) & -F^{22} F^{33} \\ -F^{11}(v^2 v_{22} \cdot v F^v + F^{11}) & -F^{22} F^{33} & \frac{F^{33}}{v_{33}}(v^2 v_{11} v_{22} \cdot v F^v + F^{11} v_{11} + F^{22} v_{22}) \end{pmatrix}$$

is positive semidefinite modulo ϕ .

Claim 3. The matrix

$$M_2 = \begin{pmatrix} \left(\frac{F^{33}}{v_{11}} + \frac{F^{11}}{v_{33}} \right) \cdot (vF^v)^2 & \frac{F^{23}}{v_{11}}(vF^v)^2 & \frac{F^{12}}{v_{33}}(vF^v)^2 \\ \frac{F^{23}}{v_{11}}(vF^v)^2 & b_{11} & b_{12} \\ \frac{F^{12}}{v_{33}}(vF^v)^2 & b_{21} & b_{22} \end{pmatrix}$$

is positive definite modulo ϕ .

Once **Claim 2** and **Claim 3** are proved, it follows that the symmetric matrix M is positive semidefinite modulo ϕ . That is, the corresponding quadratic form $f(X)$ satisfies

$$f(X) = X^T M X \gtrsim 0.$$

Combining this with formula (4.19), we conclude the proof of Theorem 4.1. $\square$

It remains to prove **Claim 2** and **Claim 3**.

Proof of Claim 2. Let us compute the leading principal minors of M_1 , denoted by $P_1(M_1)$, $P_2(M_1)$ and $P_3(M_1)$.

- $P_1(M_1)$. It is evident that

$$P_1(M_1) = \frac{F^{11}}{v_{11}}(v^2 v_{22} v_{33} \cdot vF^v + F^{22} v_{22} + F^{33} v_{33}) > 0.$$

- $P_2(M_1)$. Note that $F^{33} = F^{11}$. By (4.16) and (4.17), we have

$$\begin{aligned} P_2(M_1) &= \frac{F^{11}}{v_{11}}(v^2 v_{22} v_{33} \cdot vF^v + F^{22} v_{22} + F^{33} v_{33}) \cdot \frac{F^{22}}{v_{22}}(F^{11} v_{11} + F^{33} v_{33}) - (-F^{11} F^{22})^2 \\ &= \frac{(F^{11})^2 F^{22}}{v_{11} v_{22}} \left((v^2 v_{22} v_{33} \cdot vF^v + F^{22} v_{22} + F^{33} v_{33}) \cdot (v_{11} + v_{33}) - F^{22} v_{11} v_{22} \right) \\ &= \frac{(F^{11})^2 F^{22}}{v_{11} v_{22}} \left(v^2 (v_{11} + v_{33}) v_{22} v_{33} \cdot vF^v + F^{22} v_{22} v_{33} + F^{33} v_{11} v_{33} + F^{33} v_{33}^2 \right) \\ &\sim \frac{(F^{11})^2 F^{22} v_{33}}{v_{11} v_{22}} (v^2 (v_{11} + v_{33}) v_{22} v_{33} \cdot vF^v + vF^v) \\ &\sim \frac{(F^{11})^2 F^{22} v_{33}}{v_{11} v_{22}} (vF^v)^2 > 0. \end{aligned}$$

- $P_3(M_1)$. To compute $\det(M_1)$, we perform elementary column operations on M_1 . Specifically, we multiply the first column by $\frac{v_{11}+v_{33}}{v_{22}}$ and add it to the second column, then multiply the first column by -1 and add it to the third column. By (4.16) and (4.17), we

obtain the transformed matrix

$$\begin{pmatrix} \frac{F^{11}}{v_{11}}(v^2 v_{22} v_{33} \cdot vF^v + F^{22} v_{22} + F^{33} v_{33}) & \frac{F^{11} v_{33}}{v_{11} v_{22}}(vF^v)^2 & -\frac{F^{11}}{v_{11}}(vF^v)^2 \\ -F^{11} F^{22} & 0 & 0 \\ -F^{11}(v^2 v_{22} \cdot vF^v + F^{11}) & -\frac{F^{11}}{v_{22}}(vF^v)^2 & \frac{F^{11}}{v_{33}}(vF^v)^2 \end{pmatrix}.$$

At this point, the second and third columns become proportional. Therefore,

$$P_3(M_1) = \det(M_1) \sim 0.$$

By Lemma 2.2, the matrix M_1 is positive semidefinite modulo ϕ . $\square$

Proof of Claim 3. By the same argument, we compute the leading principal minors of M_2 , denoted by $P_1(M_2)$, $P_2(M_2)$ and $P_3(M_2)$.

• $P_1(M_2)$. Clearly,

$$P_1(M_2) = \left(\frac{F^{33}}{v_{11}} + \frac{F^{11}}{v_{33}} \right) \cdot (vF^v)^2 > 0.$$

• $P_2(M_2)$. Recall from (2.9) that

$$(4.20) \quad F^{11} F^{22} - (F^{12})^2 - (F^{23})^2 = v^2.$$

Using equations (4.16) and (4.17), we obtain

$$\begin{aligned} P_2(M_2) &= \left(\frac{F^{33}}{v_{11}} + \frac{F^{11}}{v_{33}} \right) \cdot (vF^v)^2 \cdot b_{11} - \left(\frac{F^{23}}{v_{11}}(vF^v)^2 \right)^2 \\ &= \left(\frac{F^{33}}{v_{11}} + \frac{F^{11}}{v_{33}} \right) \cdot (vF^v)^2 \cdot \left(\left(\frac{F^{11}}{v_{22}} + \frac{F^{22}}{v_{11}} - v^2 \right) \cdot (vF^v)^2 \right. \\ &\quad \left. - \left(\frac{v^2 v_{22} v_{33} + 1}{F^{11} v_{11}} + \frac{1}{F^{22} v_{22}} \right) (F^{12})^2 \cdot vF^v \right) - \left(\frac{F^{23}}{v_{11}}(vF^v)^2 \right)^2 \\ &= \frac{(v_{11} + v_{33})F^{11}}{v_{11} v_{33}} \frac{(vF^v)^3}{F^{11} v_{11} F^{22} v_{22}} \cdot \left((F^{11} v_{11} + F^{22} v_{22} - v^2 v_{11} v_{22}) \cdot F^{11} F^{22} \cdot vF^v \right. \\ &\quad \left. - ((v^2 v_{22} v_{33} + 1) \cdot F^{22} v_{22} + F^{11} v_{11}) \cdot (F^{12})^2 \right) - \frac{(F^{23})^2}{v_{11}^2} \cdot (vF^v)^4 \\ &\sim \frac{(v_{11} + v_{33})F^{11}}{v_{11} v_{33}} \frac{(vF^v)^3}{F^{11} v_{11} F^{22} v_{22}} \cdot \left((v^2 v_{22} v_{33} + 1 - F^{33} v_{33}) \cdot F^{11} F^{22} \cdot vF^v \right. \\ &\quad \left. - (v^2 v_{22} v_{33} \cdot F^{22} v_{22} + vF^v - F^{33} v_{33}) \cdot (F^{12})^2 \right) - \frac{(F^{23})^2}{v_{11}^2} \cdot (vF^v)^4 \\ &= \frac{F^{11}}{v_{11}^2} \cdot v^2 (v_{11} + v_{33}) \cdot (vF^v)^4 + \frac{v_{11} + v_{33}}{v_{11} v_{33}} \frac{F^{11}}{F^{11} v_{11} F^{22} v_{22}} (F^{11} F^{22} - (F^{12})^2) \cdot (vF^v)^4 \end{aligned}$$

$$\begin{aligned}
& - \frac{v_{11} + v_{33}}{v_{11}^2 v_{22}} F^{11} F^{33} \cdot (vF^v)^4 - \frac{1}{v_{11}^2} \cdot v^2 (v_{11} + v_{33}) v_{22} \cdot (F^{12})^2 \cdot (vF^v)^3 \\
& + \frac{v_{11} + v_{33}}{v_{11}} \frac{F^{11} F^{33}}{F^{11} v_{11} F^{22} v_{22}} \cdot (F^{12})^2 \cdot (vF^v)^3 - \frac{1}{v_{11}^2} (F^{23})^2 \cdot (vF^v)^4.
\end{aligned}$$

We continue to simplify $P_2(M_2)$. Using equations (4.16) and (4.17) again, we have

$$\begin{aligned}
P_2(M_2) & \sim \frac{F^{11}}{v_{11}^2} \cdot \frac{vF^v - 1}{v_{22}} \cdot (vF^v)^4 + \frac{v_{11} + v_{33}}{v_{11}^2 v_{22} v_{33} F^{22}} (F^{11} F^{22} - (F^{12})^2) \cdot (vF^v)^4 \\
& - \frac{vF^v - F^{22} v_{22}}{v_{11}^2 v_{22}} F^{11} \cdot (vF^v)^4 - \frac{1}{v_{11}^2} \cdot (vF^v - 1) \cdot (F^{12})^2 \cdot (vF^v)^3 \\
& + \frac{vF^v - F^{22} v_{22}}{v_{11}^2 v_{22} F^{22}} \cdot (F^{12})^2 \cdot (vF^v)^3 - \frac{1}{v_{11}^2} (F^{23})^2 \cdot (vF^v)^4 \\
& = - \frac{1}{v_{11}^2 v_{22} F^{22}} (F^{11} F^{22} - (F^{12})^2) \cdot (vF^v)^4 + \frac{v_{11} + v_{33}}{v_{11}^2 v_{22} v_{33} F^{22}} (F^{11} F^{22} - (F^{12})^2) \cdot (vF^v)^4 \\
& + \frac{1}{v_{11}^2} (F^{11} F^{22} - (F^{12})^2 - (F^{23})^2) \cdot (vF^v)^4 \\
& = \frac{1}{v_{11} v_{22} v_{33} F^{22}} (F^{11} F^{22} - (F^{12})^2) \cdot (vF^v)^4 + \frac{1}{v_{11}^2} (F^{11} F^{22} - (F^{12})^2 - (F^{23})^2) \cdot (vF^v)^4.
\end{aligned}$$

From (4.20), it follows that

$$P_2(M_2) \sim \frac{v^2 + (F^{23})^2}{v_{11} v_{22} v_{33} F^{22}} \cdot (vF^v)^4 + \frac{v^2}{v_{11}^2} \cdot (vF^v)^4 > 0.$$

• $P_3(M_2)$. Let us compute the determinant of M_2 . As before, we perform elementary column operations on M_2 , which do not change the determinant of M_2 . Specifically, we multiply the first column by $-\frac{F^{23}}{F^{11}} \frac{v_{33}}{v_{11} + v_{33}}$ and add it to the second column, then multiply the first column by $-\frac{F^{12}}{F^{11}} \frac{v_{11}}{v_{11} + v_{33}}$ and add it to the third column. These operations yield the transformed matrix

$$\begin{pmatrix}
\frac{(v_{11} + v_{33}) F^{11}}{v_{11} v_{33}} \cdot (vF^v)^2 & 0 & 0 \\
\frac{F^{23}}{v_{11}} \cdot (vF^v)^2 & b_{11} - \frac{v_{33}}{v_{11} + v_{33}} \frac{(F^{23})^2}{F^{11} v_{11}} \cdot (vF^v)^2 & b_{12} - \frac{1}{v_{11} + v_{33}} \frac{F^{12} F^{23}}{F^{11}} \cdot (vF^v)^2 \\
\frac{F^{12}}{v_{33}} (vF^v)^2 & b_{21} - \frac{1}{v_{11} + v_{33}} \frac{F^{12} F^{23}}{F^{11}} \cdot (vF^v)^2 & b_{22} - \frac{v_{11}}{v_{11} + v_{33}} \frac{(F^{12})^2}{F^{33} v_{33}} \cdot (vF^v)^2
\end{pmatrix}.$$

Note that $F^{11} = F^{33}$. By (4.16), (4.17) and (4.20), we have

$$\begin{aligned}
& b_{11} - \frac{v_{33}}{v_{11} + v_{33}} \frac{(F^{23})^2}{F^{11} v_{11}} \cdot (vF^v)^2 \\
& = \left(\frac{F^{11}}{v_{22}} + \frac{F^{22}}{v_{11}} - v^2 \right) \cdot (vF^v)^2 - \left(\frac{v^2 v_{22} v_{33} + 1}{F^{11} v_{11}} + \frac{1}{F^{22} v_{22}} \right) (F^{12})^2 \cdot vF^v
\end{aligned}$$

$$\begin{aligned}
& - \frac{v_{33}}{v_{11} + v_{33}} \frac{(F^{23})^2}{F^{11}v_{11}} \cdot (vF^v)^2 \\
& \sim \frac{vF^v}{(v_{11} + v_{33})F^{11}v_{11}F^{22}v_{22}} \left((-F^{33}v_{33} + v^2v_{22}v_{33} + 1) \cdot (v_{11} + v_{33}) \cdot F^{11}F^{22} \cdot vF^v \right. \\
& \quad - (v^2v_{22}v_{33} \cdot F^{22}v_{22} + vF^v - F^{33}v_{33}) \cdot (v_{11} + v_{33}) \cdot (F^{12})^2 \\
& \quad \left. - v_{22}v_{33}F^{22} \cdot (F^{23})^2 \cdot vF^v \right) \\
& = \frac{vF^v}{(v_{11} + v_{33})F^{11}v_{11}F^{22}v_{22}} \left(-F^{33}v_{33} \cdot F^{11}(v_{11} + v_{33}) \cdot F^{22} \cdot vF^v \right. \\
& \quad + v_{33} \cdot v^2(v_{11} + v_{33})v_{22} \cdot F^{11}F^{22} \cdot vF^v \\
& \quad + (v_{11} + v_{33}) \cdot F^{11}F^{22} \cdot vF^v \\
& \quad - v_{22}v_{33}F^{22} \cdot v^2(v_{11} + v_{33})v_{22} \cdot (F^{12})^2 \\
& \quad - (v_{11} + v_{33}) \cdot (F^{12})^2 \cdot vF^v + v_{33} \cdot F^{33}(v_{11} + v_{33}) \cdot (F^{12})^2 \\
& \quad \left. - v_{22}v_{33}F^{22} \cdot (F^{23})^2 \cdot vF^v \right) \\
& \sim \frac{vF^v}{(v_{11} + v_{33})F^{11}v_{11}F^{22}v_{22}} \left(-F^{33}v_{33} \cdot (vF^v - F^{22}v_{22}) \cdot F^{22} \cdot vF^v \right. \\
& \quad + v_{33} \cdot (vF^v - 1) \cdot F^{11}F^{22} \cdot vF^v \\
& \quad + (v_{11} + v_{33}) \cdot (F^{11}F^{22} - (F^{12})^2) \cdot vF^v \\
& \quad - v_{22}v_{33}F^{22} \cdot (vF^v - 1) \cdot (F^{12})^2 + v_{33} \cdot (vF^v - F^{22}v_{22}) \cdot (F^{12})^2 \\
& \quad \left. - v_{22}v_{33}F^{22} \cdot (F^{23})^2 \cdot vF^v \right) \\
& = \frac{(vF^v)^2}{(v_{11} + v_{33})F^{11}v_{11}F^{22}v_{22}} \left(v_{22}v_{33}F^{22} \cdot (F^{11}F^{22} - (F^{12})^2 - (F^{23})^2) + v_{11} \cdot (F^{11}F^{22} - (F^{12})^2) \right) \\
& \sim \frac{(vF^v)^2}{(v_{11} + v_{33})F^{11}v_{11}F^{22}v_{22}} \left(v^2v_{22}v_{33}F^{22} + v_{11}(v^2 + (F^{23})^2) \right).
\end{aligned}$$

Analogously, we have the following simplifications

$$\begin{aligned}
& b_{12} - \frac{1}{v_{11} + v_{33}} \frac{F^{12}F^{23}}{F^{11}} \cdot (vF^v)^2 = b_{21} - \frac{1}{v_{11} + v_{33}} \frac{F^{12}F^{23}}{F^{11}} \cdot (vF^v)^2 \\
& = F^{12}F^{23} \left(\frac{v^2v_{22}}{F^{11}} - \frac{1}{F^{22}v_{22}} \right) \cdot vF^v - \frac{1}{v_{11} + v_{33}} \frac{F^{12}F^{23}}{F^{11}} \cdot (vF^v)^2 \\
& = \frac{F^{12}F^{23}}{(v_{11} + v_{33})v_{22}F^{11}F^{22}} \cdot vF^v \cdot \left(v^2(v_{11} + v_{33})v_{22} \cdot F^{22}v_{22} - F^{11}(v_{11} + v_{33}) - F^{22}v_{22} \cdot vF^v \right) \\
& \sim \frac{F^{12}F^{23}}{(v_{11} + v_{33})v_{22}F^{11}F^{22}} \cdot vF^v \cdot \left((vF^v - 1) \cdot F^{22}v_{22} - F^{11}(v_{11} + v_{33}) - F^{22}v_{22} \cdot vF^v \right)
\end{aligned}$$

$$\sim - \frac{F^{12}F^{23}}{(v_{11} + v_{33})v_{22}F^{11}F^{22}} \cdot (vF^v)^2,$$

and

$$\begin{aligned} & b_{22} - \frac{v_{11}}{v_{11} + v_{33}} \frac{(F^{12})^2}{F^{33}v_{33}} \cdot (vF^v)^2 \\ & \sim \frac{(vF^v)^2}{(v_{11} + v_{33})F^{22}v_{22}F^{33}v_{33}} \left(v^2v_{11}v_{22}F^{22} + v_{33}(v^2 + (F^{12})^2) \right). \end{aligned}$$

Thus,

$$\begin{aligned} P_3(M_2) = \det(M_2) & \sim \frac{(v_{11} + v_{33})F^{11}}{v_{11}v_{33}} \cdot (vF^v)^2 \\ & \cdot \left(\frac{(vF^v)^4}{(v_{11} + v_{33})^2 F^{11}v_{11}(F^{22}v_{22})^2 F^{33}v_{33}} \right. \\ & \quad \cdot (v^2v_{22}v_{33}F^{22} + v_{11}(v^2 + (F^{23})^2)(v^2v_{11}v_{22}F^{22} + v_{33}(v^2 + (F^{12})^2))) \\ & \quad \left. - \frac{(vF^v)^4}{(v_{11} + v_{33})^2 v_{22}^2 (F^{11}F^{22})^2} (F^{12})^2 (F^{23})^2 \right) \\ & = \frac{(v_{11} + v_{33})F^{11}}{v_{11}v_{33}} \cdot (vF^v)^2 \cdot \frac{(vF^v)^4}{(v_{11} + v_{33})^2 F^{11}v_{11}(F^{22}v_{22})^2 F^{33}v_{33}} \\ & \quad \cdot \left((v^2v_{22}v_{33}F^{22} + v_{11}(v^2 + (F^{23})^2)(v^2v_{11}v_{22}F^{22} + v_{33}(v^2 + (F^{12})^2))) \right. \\ & \quad \left. - v_{11}v_{33}(F^{12})^2 (F^{23})^2 \right) \\ & = \frac{(vF^v)^6}{(v_{11} + v_{33})(v_{11}v_{22}v_{33})^2 F^{11}(F^{22})^2} \\ & \quad \cdot \left(v^4v_{11}v_{33} \cdot (F^{22}v_{22})^2 + v^2v_{33}^2 \cdot F^{22}v_{22} \cdot (v^2 + (F^{12})^2) \right. \\ & \quad \left. + v^2v_{11}^2 \cdot F^{22}v_{22} \cdot (v^2 + (F^{23})^2) + v^4v_{11}v_{33} + v^2v_{11}v_{33} \cdot ((F^{12})^2 + (F^{23})^2) \right) \\ & > 0. \end{aligned}$$

Therefore, the matrix M_2 is positive definite modulo ϕ . $\square$

5. PROOF OF THEOREM 1.1

In this section, we prove Theorem 1.1 by the continuity method. As a first step, we apply Theorems 3.7 and 4.1 to establish the constant rank theorem.

Theorem 5.1 (Constant rank theorem). *Let $\Omega \subset \mathbb{R}^4$ be a domain, and let $v \in C^4(\Omega)$ be a convex solution of (2.6). Then the Hessian $\nabla^2 v$ has constant rank throughout Ω .*

Proof. Define

$$l = \min_{x \in \Omega} \text{rank}(\nabla^2 v(x)).$$

As discussed in Section 1.3, it suffices to consider the cases $l = 2$ and $l = 3$. By the rank identity in Lemma 2.1, we are reduced to proving that the matrix K has constant rank $l - 2$ throughout Ω . Define

$$\Omega^* = \{x \in \Omega : \text{rank}(K(x)) = l - 2\}.$$

Since l is the minimal rank of $\nabla^2 v$ in Ω , the set Ω^* is nonempty. Moreover, Ω^* is relatively closed in Ω . On the other hand, Theorems 3.7 and 4.1, together with the strong maximum principle, imply that Ω^* is open in Ω for $l = 2$ and $l = 3$, respectively. Since Ω is connected, it follows that $\Omega^* = \Omega$. Hence, K has constant rank $l - 2$ throughout Ω . Lemma 2.1 then shows that $\nabla^2 v$ has constant rank l throughout Ω . $\square$

With Theorem 5.1 just established, it is well known that the proof of Theorem 1.1 follows standard arguments; see, for example, Caffarelli and Friedman [10], as well as [34, 37, 39, 49].

Before proceeding, we require the convexity estimates near the boundary for the function $v = -\sqrt{-u/2}$, where u satisfies the Dirichlet problem (1.3). In the following proposition, we take $\alpha = 1/2$.

Proposition 5.2 (Convexity estimates near the boundary, see Caffarelli and Friedman [10], Korevaar [33]). *Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth, strictly convex domain, and let $u \in C^2(\bar{\Omega})$ satisfy*

$$u < 0 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad |\nabla u| > 0 \quad \text{on } \partial\Omega.$$

Then there exists $\varepsilon > 0$ such that the function $v = -\log(-u)$ or $v = -(-u)^\alpha$ ($0 < \alpha < 1$) is strictly convex in the boundary strip $\Omega \setminus \bar{\Omega}_\varepsilon$, where

$$\Omega_\varepsilon = \{x \in \Omega : d(x, \partial\Omega) > \varepsilon\}.$$

We now move on to the proof of Theorem 1.1.

Proof of Theorem 1.1. If $\Omega_0 = B_1$, the unit ball in $\mathbb{R}^4$, then the unique solution to the transformed Dirichlet problem (2.7) is

$$v_0(x) = -\sqrt{\frac{1 - |z_1|^2 - |z_2|^2}{2}} \quad \text{for } x \in B_1,$$

where $x = (x_1, x_2, x_3, x_4)$, $z_1 = x_1 + \sqrt{-1}x_3$, $z_2 = x_2 + \sqrt{-1}x_4$, $|z_1|^2 = x_1^2 + x_3^2$ and $|z_2|^2 = x_2^2 + x_4^2$. Clearly, the function v_0 is strictly convex in B_1 . For an arbitrary bounded, smooth, strictly convex domain $\Omega_1 = \Omega$, define

$$\Omega_t = (1 - t)B_1 + t\Omega \quad \text{for } t \in (0, 1).$$

By the theory of convex bodies, the family $\{\Omega_t\}_{t \in (0,1)}$ provides a smooth deformation of B_1 into Ω , preserving boundedness, smoothness, and strict convexity. For further details, see Schneider [45].

For each $t \in (0, 1)$, let v_t be the solution of the Dirichlet problem

$$\begin{cases} F(\nabla^2 v_t, \nabla v_t, v_t) = 0 & \text{in } \Omega_t, \\ v_t = 0 & \text{on } \partial\Omega_t, \end{cases}$$

where

$$\begin{aligned} F(\nabla^2 v, \nabla v, v) = & v^2 [(v_{11} + v_{33})(v_{22} + v_{44}) - (v_{12} + v_{34})(v_{21} + v_{43}) - (v_{14} - v_{32})(v_{41} - v_{23})] \\ & + v [(v_2^2 + v_4^2)(v_{11} + v_{33}) + (v_1^2 + v_3^2)(v_{22} + v_{44}) \\ & - (v_1 v_2 + v_3 v_4)(v_{12} + v_{34} + v_{21} + v_{43}) \\ & - (v_1 v_4 - v_2 v_3)(v_{14} - v_{32} + v_{41} - v_{23})] - 1. \end{aligned}$$

According to the a priori estimates for the Dirichlet problem of the complex Monge-Ampère equation established by Caffarelli, Kohn, Nirenberg and Spruck [12], we have uniform bounds for $\|v_t\|_{C_{\text{loc}}^3(\Omega_t)}$, depending only on the geometry of Ω and independent of t . On the one hand, since v_0 is strictly convex, it follows that v_t remains strictly convex for all $t \in (0, \delta)$ for some sufficiently small $\delta > 0$. On the other hand, if v_t is strictly convex in Ω_t for all $t \in (0, t^*)$ with $t^* \in (0, 1)$, then v_{t^*} must be convex in Ω_{t^*} .

Now, suppose that v_1 is not strictly convex in Ω_1 . Then there exists $t_0 \in (0, 1)$ at which v_{t_0} is convex but not strictly convex for the first time. By Theorem 5.1 (Constant rank theorem), the Hessian $\nabla^2 v_{t_0}$ is degenerate at every point of Ω_{t_0} . However, Proposition 5.2 (Convexity estimates near the boundary) implies that $\nabla^2 v_{t_0}$ is of full rank in the boundary strip $\Omega_{t_0} \setminus \bar{\Omega}_{t_0, \varepsilon}$ for some $\varepsilon > 0$, which depends only on the geometry of $\partial\Omega_1$. This leads to a contradiction. Therefore, v_1 must be strictly convex in Ω_1 . Thus, we complete the proof of Theorem 1.1. $\square$

Remark 5.3. We provide the following example to demonstrate that the convexity exponent $1/2$ cannot be improved in the general class of convex domains. Similar examples can be found in Kennington [31], Ma and Xu [39], and in the recent work of Zhang and Zhou [49].

Consider the infinite open cone

$$\Gamma = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : \sqrt{(x_1)^2 + (x_2)^2 + (x_3)^2} < ax_4\},$$

where $a \in (0, 1)$. Let $\Omega \subset \Gamma$ be a bounded convex domain, with $y = (0, 0, 0, 0) \in \partial\Omega$ and $z = (0, 0, 0, 1) \in \Omega$. Define the function

$$w(x) = \frac{A}{2}(x_1^2 + x_2^2 + x_3^2 - a^2 x_4^2),$$

where $A > 0$ is chosen such that $2A^2(1 - a^2) = 16$. Thus, the function w satisfies

$$\begin{cases} (w_{11} + w_{33})(w_{22} + w_{44}) - (w_{12} + w_{34})(w_{21} + w_{43}) - (w_{14} - w_{32})(w_{41} - w_{23}) = 16 & \text{in } \Omega, \\ w \leq 0 & \text{on } \partial\Omega. \end{cases}$$

Suppose that u solves the Dirichlet problem

$$\begin{cases} (u_{11} + u_{33})(u_{22} + u_{44}) - (u_{12} + u_{34})(u_{21} + u_{43}) - (u_{14} - u_{32})(u_{41} - u_{23}) = 16 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

By the comparison principle, we have $w \leq u$ in Ω . In particular, for $t \in (0, 1)$, it follows that $-Aa^2t^2/2 = w(tz) \leq u(tz) \leq 0$. This implies $\lim_{t \rightarrow 0+} t^{-1/\alpha}u(tz) = 0$ provided $\alpha > 1/2$. Consequently, this shows that $-(-u)^\alpha$ is not convex when $\alpha > 1/2$.

In fact, if $-(-u)^\alpha$ is convex for some $\alpha > 1/2$, then we have

$$-(-u)^\alpha((1-t)y + tz) \leq -(1-t)(-u)^\alpha(y) - t(-u)^\alpha(z) = -t(-u)^\alpha(z).$$

Rearranging this inequality yields $t^{-1/\alpha}u(tz) \leq u(z) < 0$, which contradicts the fact that $\lim_{t \rightarrow 0+} t^{-1/\alpha}u(tz) = 0$. This confirms the sharpness of the convexity exponent $1/2$.

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